

**SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING
BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE**

Undergraduate Capstone Project
Submitted to the Faculty of the
Cavite State University – Carmona Campus
Carmona, Cavite

In partial fulfillment
of the requirements for the degree of
Bachelor of Science in Information Technology

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January 2026

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ACKNOWLEDGMENT

Sincerest gratitude is extended by the researchers to the following individuals for their guidance and support from the beginning until the completion of this paper:

Dr. Mark Anthony R. Abril and Mr. Marvin N. Dalisay, their thesis adviser and technical critic, for the guidance, support, consideration, and, feedbacks.

Ms. Janine B. Bacosmo, the Department Research Coordinator, for their support and guidance, their dedication, timely communication, and assistance in facilitating the necessary approvals, resources.

Dr. Regene G. Hernandez, the Department Head, for their support and encouragement, their guidance and commitment to academic excellence.

Dr. Carlo Emil B. Mañabo, Campus Research Coordinator, for his vital role in the completion of this study through his support, guidance, and commitment to upholding research standards.

Dr. Jenny Beb F. Espineli, Campus Administrator, for her support and encouragement.

Toplis Solutions Inc., their client, for the trust and participation by providing the researchers valuable information to complete the content of the system and this paper;

To their college instructors, for providing the researchers the knowledge and skills that they needed for this project;

To their family, friends, and classmates, for supporting and encouraging the researchers during their tough times, providing them with their financial and moral support;

Lastly, to the Almighty God, for being the researchers' source of strength, wisdom, courage, intelligence, and determination to finish this paper.

THE RESEARCHERS

ABSTRACT

ABUEDO, MICHAEL JHEY P., CABALLES MARK VINCENT T., LANGUITA MEINARD C. and RODRIGUEZ, LORENCE R., SecureAttend: An Attendance Management System Using Biometrics for Toplis Solutions Inc., in Silang, Cavite. Undergraduate Capstone Project. Bachelor of Science in information Technology. Cavite State University — Carmona Campus, Carmona, Cavite. March 2026. Adviser: Dr. Mark Anthony R. Abril.

Toplis Solutions Inc., located in Silang, Cavite, relied on manual attendance monitoring through daily log sheets, which led to issues such as human error, delayed attendance validation, and the possibility of buddy punching. These challenges made it difficult for administrators to accurately monitor employee attendance and manage workforce records efficiently. This study aimed to develop SecureAttend: An Attendance Management System Using Biometrics to improve the organization's attendance monitoring process and provide a more reliable and automated solution for recording employee attendance. The system streamlined attendance tracking through fingerprint biometric authentication, employee management, schedule management, department management, notification alerts, and automated attendance report generation. SecureAttend was developed using the Modified Waterfall Model, which included the phases of requirement analysis, system design, implementation, system testing, and deployment and maintenance. The system was implemented using modern web development technologies including React.js, Node.js, Express.js, MySQL, JavaScript, and C# for biometric integration. The system was evaluated using the ISO/IEC 25010 software quality model, with responses gathered from 80 respondents composed of IT specialists, domain experts, the organization administrator, and employees. Evaluation results indicated that the system achieved favorable ratings across product quality characteristics including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Additionally, the quality in use evaluation obtained excellent results in effectiveness, efficiency, satisfaction, freedom from risk, and context coverage. These

findings confirmed that the SecureAttend system effectively improved attendance monitoring, enhanced accuracy and transparency, and supported efficient workforce management within the organization.

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An undergraduate capstone project proposal submitted to the faculty of the Department of Industrial and Information Technology, Cavite State University, Carmona Campus, Carmona, Cavite in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology. Prepared under the supervision of Dr. Mark Anthony R. Abril

INTRODUCTION

Staffing agencies play an important role in providing employers with diverse workforce solutions across various industries. They serve as a bridge between companies and workers. They do this by ensuring that organizations have access to qualified personnel whenever they are needed whether it is for temporary assignment, project-based roles, or for long term employment. By managing the recruitment, placement, and coordination of workers, staffing agencies support organizations in maintaining productivity and adapting to the ever-changing workforce demands.

Project Context

Toplis Solutions Inc. was a service provider company established in 2004. The company began its operations as a primarily logistics service provider but expanded over the years to sourcing, recruiting, and deploying top talents aligned with the unique needs of its clients. Toplis Solutions Inc., located in Maguyam, Silang, Cavite, was responsible for providing staffing services for Alfamart branches near its location.

Toplis Solutions Inc. revealed several operational issues, one of which was warehouse overcrowding. This situation occurred when some employees scheduled

for a shift arrived late. Employees were given a grace period of one hour before the start of the shift; however, if they failed to arrive within this period, they were automatically marked absent and replaced by buffer employees, who were standby personnel ready to take their place. The problem arose when these late employees were not informed beforehand by their team leader that they had already been replaced. As a result, they could still enter the warehouse premises, causing the number of employees to exceed the maximum capacity of the warehouse. Furthermore, the manual process of attendance validation using attendance sheets required the team leader to manually verify which employees were not supposed to report for that shift.

Another issue identified was the manual attendance process of the employees. Employees filled out an attendance logbook and used a document called the Daily Log Sheet (DLS), where they recorded their personal information, including the time and type of action performed, such as clocking in or clocking out. The DLS was then signed by the security guard for validation, who was responsible for recording the exact time of the employees' arrival at the warehouse. After validation, the DLS was returned to the employee for safekeeping. At the end of the shift, all DLS forms were dropped into a designated collection box. These were then collected by the team leader, who verified that the number of DLS forms matched the number of entries in the attendance logbook. After validation, the DLS forms were forwarded to the Human Resources department for encoding. The Human Resources personnel manually encoded each employee's information, including employee ID, name, shift, and work hours.

The inefficient scheduling process, the manual method of attendance tracking, and the high administrative workload indicated the organization's need for a more reliable and automated system. As a solution, SecureAttend was proposed as a fingerprint biometric-based attendance system connected to a web platform. The system ensured accurate and tamper-proof attendance tracking, streamlined employee scheduling,

and reduced administrative efforts by automating attendance validation and report generation processes.

These issues were aligned with several Sustainable Development Goals (SDGs), specifically SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 16 (Peace, Justice, and Strong Institutions).

Objectives of the Study

The primary objective of the study was to develop a web-connected attendance management system that utilized fingerprint biometrics and helped in attendance tracking, employee scheduling, and reducing manual labor in attendance validation and encoding.

Specifically, it aims to:

1. design a SecureAttend that:
 - a. enroll employees using fingerprint biometric data;
 - b. tracks attendance record using fingerprint authentication;
 - c. validates employee attendance verification through fingerprint;
 - d. manage employee scheduling;
 - e. provides notification and alerts;
 - f. generates attendance reports; and
 - g. allows system configuration and modification
2. developed web application using the following:
 - a. JavaScript for the primary programming language;
 - b. C# for the secondary programming language
 - c. MySQL for the database;
 - d. React.js for frontend framework;
 - e. Tailwind CSS for the CSS framework;
 - f. Node.js for JavaScript runtime environment;
 - g. Express.js for web application framework;
 - h. Figma for the design tool; and

- i. Visual Studio Code for the development environment.
- 3. tested the system in terms of unit, integration, and system testing;
- 4. evaluated the system using the adopted International Organization for Standardization (ISO) 25010 evaluation instrument; and
- 5. prepared an implementation plan.

Purpose and Description

The primary purpose of this research was to develop a fingerprint biometric web-connected attendance management system for Toplis Solutions Inc. in Maguyam, Silang, Cavite that addressed issues related to warehouse overcrowding, the existing manual attendance process, and the manual validation and encoding procedures. The developed system aimed to enhance attendance tracking and employee scheduling by providing accurate attendance data, streamlining shift management, and minimizing manual administrative work in attendance validation and encoding.

The primary beneficiaries of SecureAttend were the different stakeholders of Toplis Solutions Inc. For the Human Resources department, the system reduced the manual effort of encoding employee details for payroll, as validated attendance records were automatically generated and linked to employee IDs. This allowed Human Resources personnel to focus on other essential tasks. Team leaders and supervisors benefited from having an overview of validated attendance within their assigned zones, as well as simplified tools for managing employee schedules. Employees also gained convenience by no longer needing to manually fill out attendance sheets. They were able to view their work schedules, monitor their attendance records, and receive notifications regarding schedule changes. Overall, SecureAttend improved attendance accuracy, streamlined scheduling and validation processes, and reduced administrative workload across Toplis Solutions Inc.

This study supported several Sustainable Development Goals (SDGs). It aligned with SDG 8 (Decent Work and Economic Growth) by promoting a fair and transparent attendance system, reducing errors and fraudulent practices, and

improving work efficiency. It also supported SDG 9 (Industry, Innovation, and Infrastructure) by utilizing technology to modernize company operations, making attendance tracking more accurate and automated. Finally, it contributed to SDG 16 (Peace, Justice, and Strong Institutions) by ensuring transparency and fairness in attendance records.

Through the developed fingerprint biometric web-connected system, Toplis Solutions Inc. was able to enhance employee scheduling management, accurately track attendance, and reduce the time and manual effort required for validation and encoding. The study addressed specific operational challenges of the company, such as warehouse overcrowding and inefficient attendance processes, while also contributing to broader efforts in promoting fair labor practices, workplace innovation, and stronger organizational policies.

Time and Place of the Study

The implementation period of the study was from February 2025 to mid-April 2026. The study was conducted at Toplis Solutions Inc., located at Brgy. Maguyam, 2/F DC Petroleum Gas Station Building, Silang, 4118 Cavite.

Scope and Limitation of the Study

This study focused on the development and evaluation of SecureAttend, a web-connected fingerprint biometric attendance system designed to automate manual tracking, validation, and scheduling processes for Toplis Solutions Inc. in Maguyam, Silang, Cavite. The system featured biometric clock-in and clock-out, schedule-based validation, automated notifications, and report generation. Role-based access allowed employees to view their records, team leaders to manage schedules and monitor zone attendance, and Human Resources personnel to administer user accounts and retrieve validated data for payroll preparation.

Conducted from February 2025 to mid-April 2026, the project was limited to Toplis Solutions Inc. and required adaptation for use in other organizations. The system did not include leave management or direct payroll integration. While a stable

internet connection was required for real-time synchronization, the biometric devices were capable of temporarily storing records during network interruptions. Minor limitations also included potential fingerprint recognition errors due to dirty scanners or worn fingerprints.

Conceptual Framework

The conceptual framework shows the understanding between the system input, process, and output.

Figure 1 shows the conceptual framework of the study.

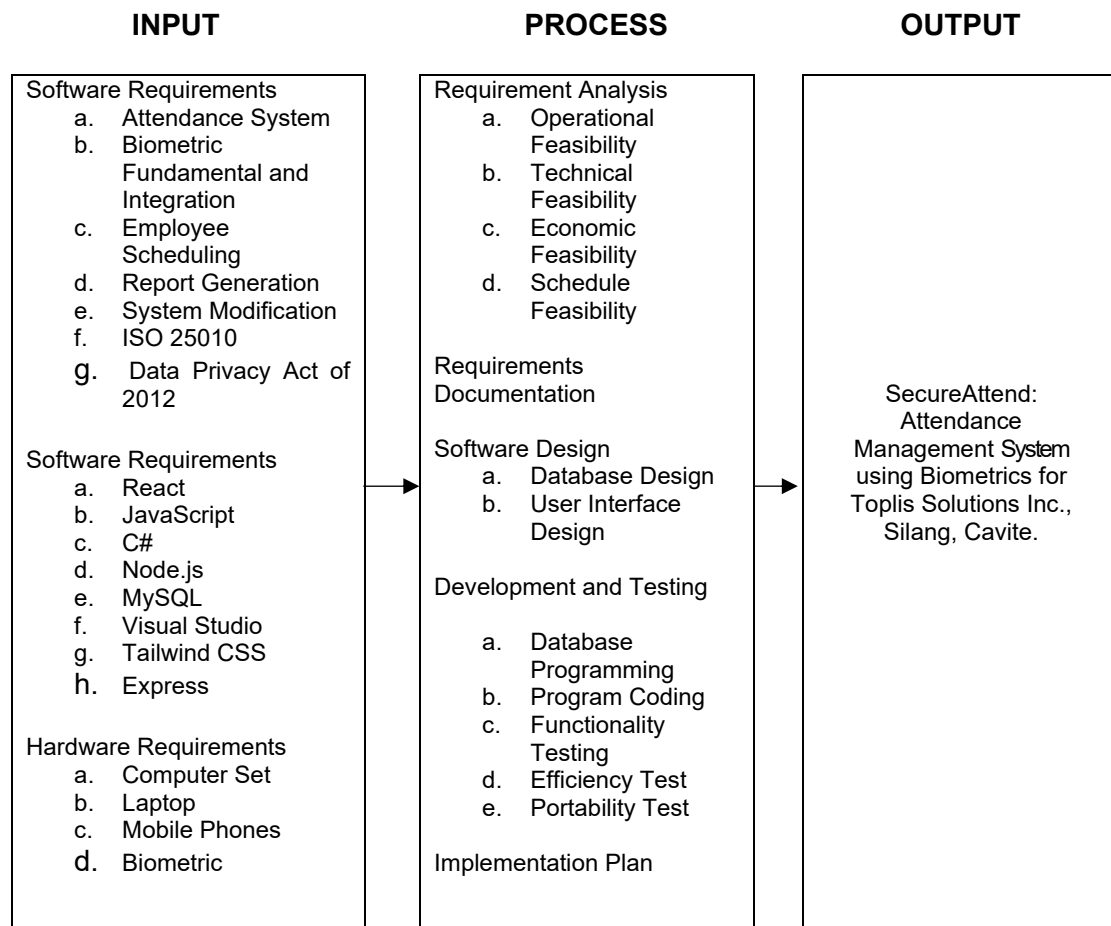


Figure. 1 Conceptual framework of the study

The input consisted of three requirements, namely: knowledge requirements, software requirements, and hardware requirements. The knowledge requirements included practical knowledge in biometric attendance systems, fundamentals and

integration, employee scheduling, report generation, system modification, ISO 25010, and the Data Privacy Act of 2012. The software requirements included the tools needed for developing the proposed project, such as React.js and CSS for front-end development, JavaScript and C# as programming languages, Visual Studio Code as the integrated development environment, MySQL for database administration, and Node.js for back-end development. The hardware requirements included laptops, desktops, and mobile phones for system access, a biometric scanner for attendance validation, and a server or web hosting service for system deployment.

Meanwhile, the process was divided into five stages: requirement analysis, requirement documentation, software design, development and testing, and implementation planning. These stages were followed in order to develop the system. Requirement analysis involved operational, technical, economic, and schedule feasibility to determine whether the proposed project was viable. Following requirement analysis, requirement documentation was conducted to describe the functional and non-functional requirements of the project. Afterward, software design was carried out, which involved designing the database structure and user interface. This was followed by development and testing, which included database programming, program coding, and various tests such as functionality, efficiency, and portability testing. This phase was considered crucial in the development of the proposed project as it evaluated the overall performance of the system. Lastly, implementation planning was conducted to ensure a seamless transition and successful user acceptance during the integration of the system into the attendance processes of Toplis Solutions Inc., Maguyam, Silang, Cavite.

The output of the conceptual framework was a fully developed SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite. The system included key features such as a centralized platform for supervisors, attendance tracking functionalities. In the final stage, the system was evaluated in terms of efficiency, usability, and reliability. The evaluation was conducted

based on user feedback, system performance testing, and in accordance with ISO 25010 software quality standards.

Definition of Terms

Biometrics are each employee's unique physical characteristics such as fingerprints which are used for identity verification.

Biometric attendance application refers to the standalone system that records employee attendance using fingerprint authentication and synchronizes the data with the SecureAttend web system.

Clock-in refers to the action performed by an employee to record the start of work using the biometric device.

Clock-out refers to the action performed by an employee to record the end of work using the biometric device.

Synchronization refers to the process of automatically sending locally stored attendance records from the biometric device to the web application once internet connectivity is available.

Validation is the process of confirming attendance data recorded in the system.

REVIEW OF RELATED LITERATURE

This chapter discussed the review of related literature and studies, which provided a foundation for understanding the research, technologies, and methodologies relevant to SecureAttend: An Attendance Management System using Biometrics for Toplis Solutions Inc. The chapter included studies related to attendance management, biometrics, real-time monitoring, web-based systems, and data security, highlighting the strengths and limitations of existing solutions. These helped in identifying the rationale for adopting a more secure, automated, and user-friendly approach to attendance management. SecureAttend addressed various issues in attendance management identified in the reviewed studies, including buddy punching, unauthorized access to attendance records, and human errors in attendance verification. The reviewed literature and studies established the theoretical and technological basis for the proposed system and supported its relevance in improving the attendance management process.

Related Literature/Studies

Attendance System. Efficient employee attendance management enables any company to improve overall corporate performance and achieve specific objectives. Accurate employee attendance records are essential for maintaining workplace discipline and increasing worker efficiency. Manual attendance-time monitoring increases the cost of time-consuming and paper-intensive business operations. Human activities, such as workplace mistakes and fraudulent timekeeping, are extra hidden costs that have an impact on the organization's efficiency (Oo *et al.*, 2018).

Traditional attendance management methods, such as manual sign-ins and RFID card-based systems, have various flaws, including security risks,

inefficiencies, and vulnerability to human mistake (Senapatha & Nendya, 2024).

Manual attendance systems, often referred to as traditional attendance tracking methods, have been the foundation of attendance tracking for many years. They rely on manual record-keeping methods, such as sign-in sheets, time punch cards, or pen-and-paper registers, where people enter their arrival and departure times (Qureshi, 2025).

Manual attendance systems, while typical provide various issues that can hinder corporate efficiency and accuracy. One key concern is the potential for human mistake; manual data entry can result in inconsistencies in attendance records, leading to payroll anomalies and staff dissatisfaction. Furthermore, behaviors such as "buddy punching," in which one employee clocks in for another, contribute to time theft and higher labor costs. Manual systems also lack real-time monitoring capabilities, making it difficult for managers to track attendance in a timely manner, particularly in remote or hybrid work environments (Faleiro, 2024).

Buddy punching, in which employees' clock in or out on behalf of their coworkers, is a common type of time theft that can have serious consequences for businesses. This practice causes financial losses, poor productivity, and a culture of dishonesty in the company (Tempus Central, 2023).

Buddy punching has negative consequences on company productivity and morale in addition to financial costs. When dishonest methods go unquestioned, they can create an unequal atmosphere, reducing motivation among honest employees and sometimes causing scheduling issues due to incorrect attendance records (Tempus Central, 2023).

Automated Attendance Systems. Today we live in the digital era, where digital systems have been utilized widely in many organizations since they are considered safe and difficult to sabotage. One type of automated AMS is a

Web-based attendance system. Every employee should be able to access the website to log in. When the business uses this kind of attendance system, employees must browse the website, enter their username and password, and clock in (Seivani, 2022).

Biometric Fundamental and Integration. A new and rapidly expanding technology that makes identity verification safe and easy to use is biometric authentication. with distinct behavioral and physiological markers. The study aims to address a theory regarding the evolution of biometric systems, risks and limitations by looking at several biometric technologies, such as fingerprint, facial recognition, iris scanning, and biometrics of behavior. By combining artificial intelligence and machine learning, these systems are now more precise and dependable, so they can be applied to border control, banks, and mobile devices. Given that biometric verification techniques are although advantageous, there are a number of disadvantages, such as privacy concerns, data breaches, spoofing attempts, and legal problems. Although these dangers are inevitable since more and more businesses are using biometric systems, thus it makes sense that data security is robust, so laws and regulations are adhered to, and that the procedure is morally right. The paper illustrates why security and user experience must be balanced. convenience that ensures biometric data is securely stored and transmitted while adhering to changing regulations. It also emphasizes how crucial it is to continuously monitor and update the biometric authentication systems. To ensure their security and integrity. Biometric authentication has the potential to be a potent instrument for enhancing security in numerous sectors. However, taking into account several technological, legal, and ethical considerations is crucial to achieve success in order to reduce whatever dangers that such a system could pose (Journal of Computer Science and Technology Studies, 2024).

Fingerprint trait is among one the most popular biometric modality. We will be using a unimodal biometric system namely an Automatic Fingerprint Identification System (AFIS) as our illustrative case. Fingerprint image can be

affected by different distortion types resulting from hardware and software device errors of the acquisition subsystems for images transmission, compression, and restoration. The negative impact of those various distortion types is reduced by the addition of an image denoising phase as reported in most books on biometrics (Bleay, Croxton, & De Puit, 2018).

The iris and retina scanning technologies store biometric data a high-resolution eye image captured by infrared light and compare it with previously stored data to ensure the security and authentication processes. While fairly accurate, iris recognition is one of the most reliable biometric methods, as the patterns can be complex and unique. Nevertheless, the main limitation of iris and retina scanning is the requirement for sophisticated, costly hardware and entrepreneurial development, which may limit their reach and applicability (Ankerman, 2016).

Recently, behavior biometrics have gained significant attention in computer vision, proving effective in engineering applications like healthcare monitoring, sports analysis, enhancing safety, and HAR (Himeur *et al.*, 2023).

The main idea of voice recognition is to identify the identity based on the unique characteristics of an individual's voice. It goes through many voice features such as pitch, cadence, accent, and dynamics of speech, including rhythm and tone (Raju, 2017).

All these factors lead to a unique vocal signature that can be used to verify the speaker. Phone systems, virtual assistants, and customer service applications are typical applications that use voice recognition (Kepuska & Bohouta, 2018).

Employee Scheduling. In several lines of work the employees have to work in different shifts to cover various needs, including working with different areas of health care, protection services, transportation, manufacturing or call centers. Nevertheless, the creation of feasible schedules is subject to many factors that have

to be addressed. The demands can be presented in different forms, compliance with various legal requirements is imperative and the workers' contentment has to be considered. Hence, automated solutions become a necessity in order to remain competitive. However, even in this case, it is oftentimes difficult to arrive at quality solutions in a reasonable time frame, as several of the issues are NP-hard. Consequently, a new framework is put forward for the generic employee scheduling problem that permits the execution of different heuristic algorithms and their employment for a broad spectrum of problems. This is achieved through the introduction of a cohesive approach to constraints and the option to execute different maneuvers that can be shared amongst different algorithms. Moreover, a novel searching technique is created and integrated into the framework (Computer and Operations Research, 2020).

Based on the analysis of various employee scheduling problems in literature, a new specification format was developed (Kletzander *et al.*, 2017) that supports a wide range of problems using different demand specifications and different types of definitions.

A new approach based on Simulated Annealing is implemented in our framework and applied to various benchmark instances from literature for comparison as well as to instances from a new instance generator. The instances from literature cover nurse rostering (Curtois, 2017) as well as different problems involving tasks from Smet *et al.* (2016) and Lapgue *et al.* (2013).

Production shifts refer to the scheduling of equipment and workers in a production process based on the job quantity. Since workers may not work continuously, they have segmented work and rest times. Thus, the shift arrangement of each worker may vary daily, adding complexity to the production process. A job scheduling plan is the chronological ordering of jobs with corresponding production resources. The job shop scheduling problem

(JSP) is widely discussed in various manufacturing fields, such as mechanical equipment (Jong *et al.*, 2020), electronics (Lin *et al.*, 2022), and aerospace (Li *et al.*, 2022a).

Employee planning has been extensively researched in many fields (Castillo-Salazar *et al.*, 2016). Despite the changing nature of tasks for production workers in machine-dominated production environments, employee planning will remain significant in the smart factory (Schwemmer *et al.*, 2022).

Report Generation. Nawi & Sahid (2024) discusses that in the information age, in all aspects of the life of the global community, the role of communication technology is very important. Aji *et al.* (2023) in recent years, it has become increasingly difficult to use manual methods that require a lot of time and effort in managing employee attendance. Therefore, to improve the efficiency and accuracy of employee attendance management, many companies have switched to online attendance systems. Amoguis *et al.*, (2021) developed an android application called Digital Attendance and Accomplishment Report Monitoring (DigiAtt) for the employees and staff of Bohol Island State University-Candijay Campus to help them in reducing the hassle in generating daily time records of each employee.

System Modification. The increase of software content in manufactured goods is being driven by the rising digitization of the physical environment and products. This calls for improved performance in product development and operational excellence to handle the increased complexity of projects, products, and systems. Even businesses with top-tier quality systems have been weakened by inability to handle the increasing complexity, which has damaged brand equity and decreased sales. Unmanaged complexity poses a risk to supply chains for manufacturing and product development, endangering the Internet of Things' (IoT) potential for growth, which depends on linked and intelligent devices. The number of components and modules will grow linearly in traditional product development organizations, but the number of potential software interactions will not grow exponentially. The complexity

brought about by the growing amount of software in conventionally hardware goods is examined from the standpoint of innovation demand (e.g., the expansion of IoT). The foundation for strategic product integration is then established by modeling the complexity of hardware/software convergence. It has been discovered that component interaction coverage offers strategic leverage for controlling the efficacy of integration (IEEE Engineering Management Review, 2021).

The Internet of Things (IoT) is a unique concept in the field of computer networking that enables the communication of various types of objects via the Internet (Fenanir *et al.*, 2019). These devices can be RFID tags, actuators, sensors, or mobile phones; all of which connect and cooperate utilizing a single addressing technique (Abbas *et al.*, 2019).

IoT technologies present various previously unimagined prospects for human interconnection. Hypertext transfer protocol, message queuing telemetry transport (MQTT), and domain name system are the internet protocol that runs IoT services under the TCP/IP architecture (Saheed, 2022).

IoT has the potential to improve people's lives and also contribute to the development of the smart city. The IoT provides a centralized stage for sharing of information across the industry, environment, and society (Singh *et al.*, 2020). As a result, efficient coordination between promised services and accessible resources is required for IoT-enabled intelligent cities to function effectively (Arshad *et al.*, 2020).

By gaining access to users' personal information, spying, and causing financial losses, attackers may jeopardize their privacy and security. Apart from compromising the security of the IoT, these intrusions pose a threat to the entire ecosystem, including applications, sites, servers, and social media via controlled smart objects such as robot networks (Li *et al.*, 2020).

ISO 25010. The development of attendance monitoring systems that use ISO/IEC 25010 standards improves software quality by emphasizing important

features including usability, security, maintainability, portability, performance efficiency, and functional applicability. For example, in order to guarantee that the system fulfilled high requirements for quality, a web-based smart attendance tracking system that used GSM modules for SMS notifications and QR codes (Bondoc, 2023). Analyzed against ISO 25010 standards, Dolor *et al.* (2021) also provided an improved attendance monitoring system that uses face recognition and QR codes, improving functionality and resolving monitoring concerns.

These studies support SecureAttend's goals of providing a safe, effective, and user-friendly solution by showcasing the real-world use of ISO/IEC 25010 in creating reliable attendance systems.

Data Privacy Act of 2012. The Philippine Data Privacy Act (DPA) of 2012 was enacted to protect the personal information of its citizens from being disclosed without its consent. The National Privacy Commission (NPC) was established in 2015 to promote, regulate, and monitor data privacy compliance of both Government and Private Institutions. This study sought to explore and explain how and why do the Philippine Government agencies comply with the DPA 2012. (American Scientific Publisher, 2018)

System Technical Background

Different software had been employed in the development of the system. The researchers chose the following software requirements: (1) ReactJS; (2) JavaScript; (3) C#; (4) Node.js; (5) MySQL (6) Visual Studio (7) Tailwind CSS (8) Express. The hardware requirements are: (1) Computer Set (2) Laptops and (3) Mobile Phones.

ReactJS. Facebook created ReactJS, a JavaScript library for developing quick, scalable, and interactive web apps (Jordan, 2020). It uses a component-based architecture, which improves code reusability and makes UI creation easier. In web-based attendance systems, ReactJS is critical for real-time data visualization, dynamic attendance record rendering, and improving user experience through seamless state management. Furthermore, ReactJS supports progressive web apps (PWAs), which

make attendance systems mobile-friendly and accessible on many platforms (Williams, 2019). Because SecureAttend seeks to deliver real-time monitoring and notifications, ReactJS's event-driven architecture enables smooth integration with APIs for MFA (Multi-Factor Authentication), device tracking, and push notifications.

JavaScript. plays a pivotal role in developing robust attendance systems like SecureAttend, especially in implementing Multi-Factor Authentication (MFA) and real-time monitoring features. The Web Authentication API (WebAuthn) enables the integration of MFA using JavaScript, enhancing security by requiring users to verify their identity through multiple factors, such as hardware security keys or biometric data. This approach significantly reduces reliance on traditional passwords and mitigates phishing risks (Sling Academy, 2024).

C#. C#, pronounced "C sharp," is a general-purpose, object-oriented programming language used to develop a wide range of programs, including enterprise software, video games, and mobile apps. Introduced by Microsoft in 2002 with its release of Visual Studio .NET, C# is part of the C family of languages, which also includes C and C++. C# is the most modern of the three and easiest to learn since it's a high-level programming language (Coursera, 2024).

Node.js. Arushi Tandon (2021) created an Attendance Management System utilizing Node.js, Express.js, and MySQL, with a focus on data accuracy and real-time tracking. The study demonstrates how to integrate RESTful APIs for seamless data retrieval and manipulation, as well as how Node.js increases the performance and efficiency of attendance monitoring systems (Tandon, 2021).

MySQL. The researchers describe the use of MySQL to create a Human Resource Management System, highlighting the need of establishing a strong database schema for efficient personnel data management. Madyatmadja and Adora (2019) demonstrate how a well-structured MySQL database can improve attendance monitoring, eliminate manual errors, and increase data accessibility.

Visual Studio Code (VS Code) is a versatile, open-source code editor developed by Microsoft, renowned for its robust features and extensive extension ecosystem, making it an ideal tool for developing secure, real-time attendance monitoring systems like SecureAttend (GitGuardian, 2025).

Tailwind CSS. Projects like this one demonstrate how Tailwind CSS can be used with technologies like Next.js and PostgreSQL to create sophisticated attendance control systems. These systems include user identification, student record management, attendance monitoring, modern dashboards, and interactive charting, demonstrating Tailwind CSS's ability to promote unified and fast user interface development by (Bhosale, 2025).

Express. Attendance Management System by Rs07-404: This system uses Express.js, EJS templates, and PostgreSQL to provide secure login, attendance management, and customizable attendance pages. The features include secure authentication, class selection for attendance management, and the ability to view and update attendance records (Rs, 2025).

Synthesis

The reviewed literature and studies collectively emphasized the growing demand for automated and secure attendance management solutions, particularly in organizations where manual and semi-automated methods were still widely used. Studies on traditional attendance systems revealed that manual sign-in sheets and card-based methods were prone to human error, time theft, and inefficient record-keeping, resulting in inaccurate attendance data and increased administrative workload (Oo et al., 2018; Faleiro, 2024; Tempus Central, 2023). These challenges were evident in many workplace environments and highlighted the necessity of transitioning to more reliable and technology-driven attendance systems.

Research on automated and web-based attendance systems demonstrated that digital platforms significantly improved the accuracy, efficiency, and accessibility

of attendance records. Web-based systems allowed employees and administrators to access attendance data in real time while reducing reliance on paper-based processes (Seivani, 2022). However, several existing systems primarily focused on basic time logging and often lacked advanced security measures or role-based access control, which limited their effectiveness in preventing fraudulent practices such as buddy punching and unauthorized attendance modification.

Studies on biometric technologies further reinforced the value of biometric authentication in attendance management. Fingerprint-based biometric systems provided a secure and non-transferable method of identity verification, ensuring that attendance records were accurately linked to individual employees (Bleay et al., 2018). Compared to other biometric modalities such as iris, retina, or voice recognition, fingerprint biometrics offered a practical balance of accuracy, cost-effectiveness, and ease of deployment, making them suitable for organizational use. While biometric systems introduced concerns related to data privacy and security, the literature emphasized the importance of secure storage, encryption, and compliance with data protection regulations to mitigate these risks (Journal of Computer Science and Technology Studies, 2024).

The literature also highlighted the importance of employee scheduling and reporting in workforce management. Automated scheduling systems helped prevent overlapping shifts, understaffing, and scheduling conflicts, which were common issues in manual scheduling processes (Kletzander et al., 2017; Computer and Operations Research, 2020). In addition, report generation modules enabled organizations to analyze attendance trends, support payroll processing, and improve decision-making. Studies on digital reporting systems confirmed that automated report generation enhanced operational efficiency and reduced errors associated with manual computation (Nawi & Sahid, 2024; Amoguis et al., 2021).

Beyond functional features, the reviewed studies underscored the importance of system quality, security, and maintainability. The application of ISO/IEC 25010 standards provided a structured approach to evaluating software quality attributes such as usability, performance efficiency, security, and maintainability. Previous attendance management systems evaluated using ISO/IEC 25010 demonstrated improved system reliability and user satisfaction when these quality standards were applied (Dolor et al., 2021; Bondoc, 2023). Compliance with data privacy regulations, particularly the Philippine Data Privacy Act of 2012, was also emphasized to ensure the ethical handling of biometric and personal data.

Overall, the reviewed literature identified persistent challenges in attendance management, including inefficient manual processes, security vulnerabilities, inaccurate records, and limited monitoring capabilities. While existing systems addressed some of these issues, many focused on isolated functionalities or relied on technologies that were not practical for all organizations. SecureAttend addressed these gaps by integrating fingerprint-based biometric authentication, web-based attendance monitoring, automated scheduling, and comprehensive reporting within a single system. By synthesizing proven concepts from prior studies and aligning them with organizational needs and quality standards, SecureAttend was positioned as a secure, efficient, and user-friendly solution for modernizing attendance management processes.

METHODOLOGY

This chapter presented the approach used to design, develop, and evaluate SecureAttend, focusing on preventing unauthorized access, buddy punching, and attendance fraud. It outlined the research design, development process, and testing methods, following a structured Software Development Life Cycle (SDLC) to ensure the system's security, usability, and reliability.

Design of Software, Systems, Product, and /or Processes

The following conceptual designs, data models, and block diagrams in this section were utilized in the system's development. Presented below is the wireframe of SecureAttend.

Figure 2 Shows the Employee Dashboard of the System.

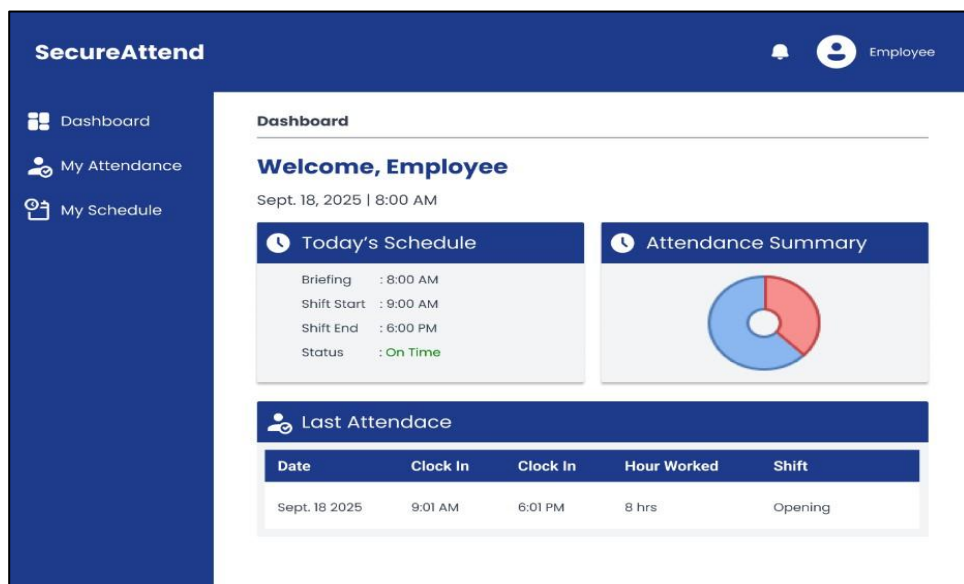


Figure 2. Wireframe of Secure Attend (Employee Dashboard)

This figure shows the Employee Dashboard. Employees can see their assigned working schedule at the top along with the current date and time. The dashboard also displays their most recent clock-in and clock-out record, including whether the log has

been synced. At the bottom, employees can view a quick summary of their attendance status to help them monitor their own records.

Figure 3 Shows the My Attendance Page of the System.

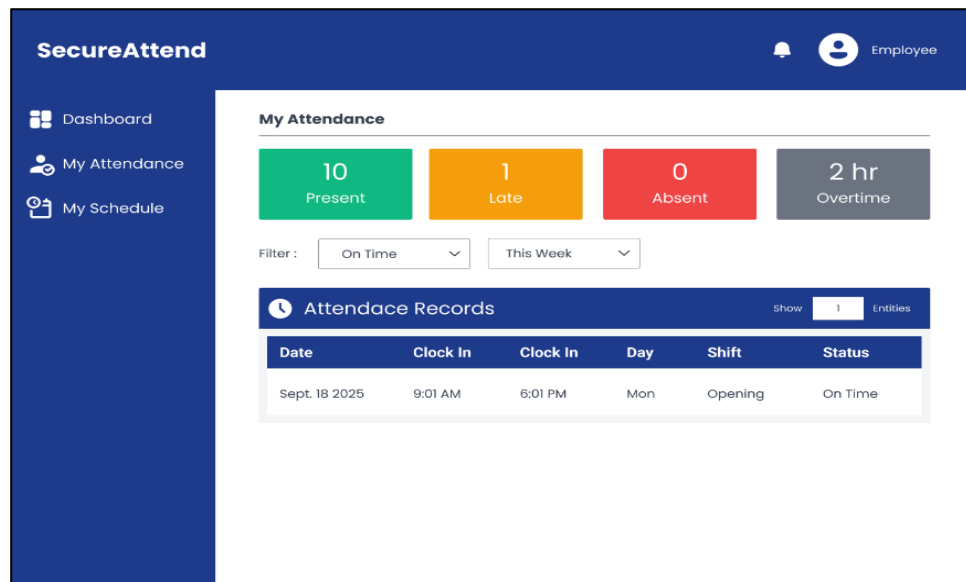


Figure 3. Wireframe of SecureAttend (My Attendance Page)

This figure shows the My Attendance page for employees. It presents a table of daily attendance logs with details such as time in, time out, and status (present, late, or absent). To make navigation easier the page include a filter option that allow employee to sort or search their records based on a specific date range or attendance status. At the bottom of the page, a summary that provide a quick overview of the employee's overall attendance performance this section highlights the total days present, absent, or late. This summary gives the user a comprehensive view of their attendance habit. This page serves as a convenient and organized tool for managing and reviewing attendance records in one place.

Figure 4 shows the My schedule of the System.

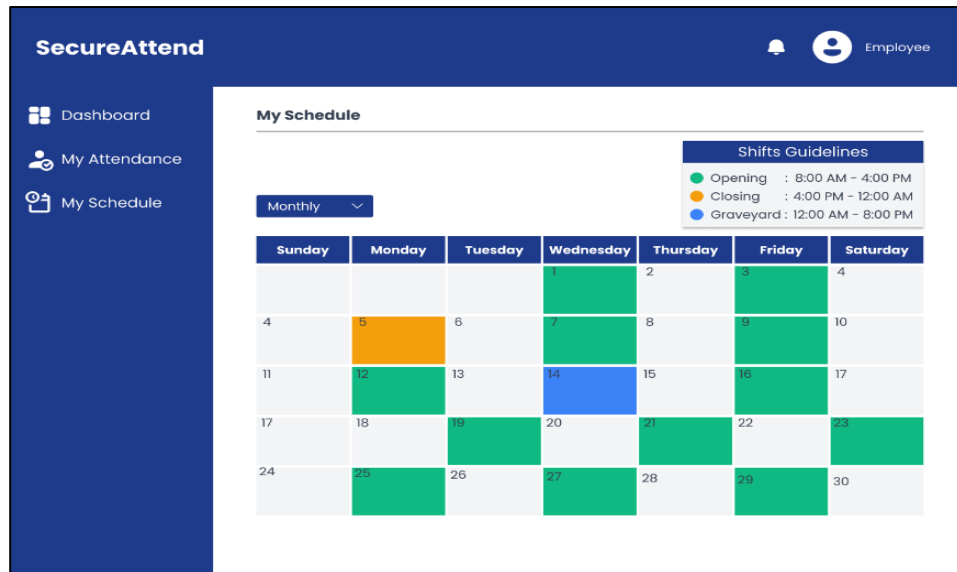


Figure 4. Wireframe of SecureAttend (My Schedule)

This figure shows the My Schedule page where employees can view their assigned shifts. The schedule is displayed either in weekly or monthly format, with each shift clearly labeled and color-coded according to shift type (morning, afternoon, or night). Rest days are also indicated, allowing employees to easily plan and prepare for their workdays.

Figure 5 shows the Team Dashboard page of the system.

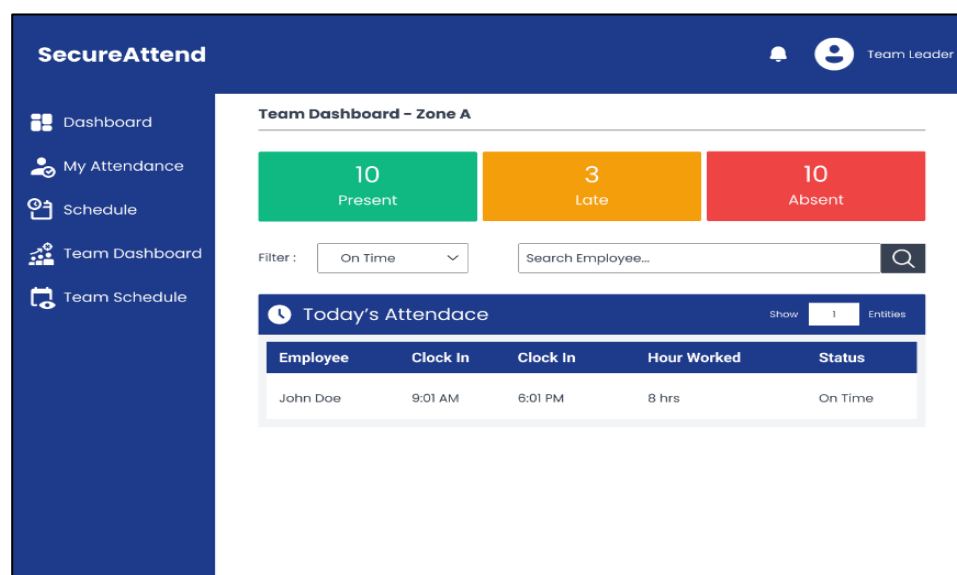


Figure 5. Wireframe of SecureAttend (Team Dashboard)

This figure shows the Team Dashboard for team leaders. It provides a real-time overview of today's attendance within their team, including the number of members present, late, or absent. A recent logs section lists the latest clock-in and clock-out activities of team members. Summary cards give quick insights into team attendance, while shortcuts allow the team leader to access schedule management and reports for their group.

Figure 6 shows the Admin Dashboard page of the system.

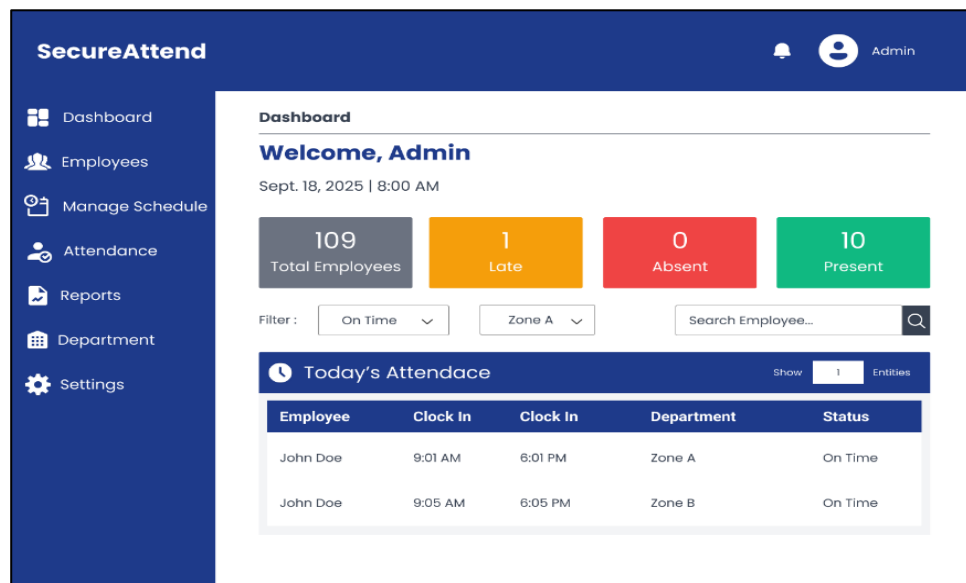


Figure 6. Wireframe of Secure Attend (Admin Dashboard)

This figure shows the Admin Dashboard. It provides administrators with an overview of today's company-wide attendance, including the number of employees present, absent, or late. A recent logs section displays the latest clock-in and clock-out activities from employees. Summary cards and charts also provide administrators with a quick understanding of workforce attendance trends.

Figure 7 shows the Manage Employees page of the system.

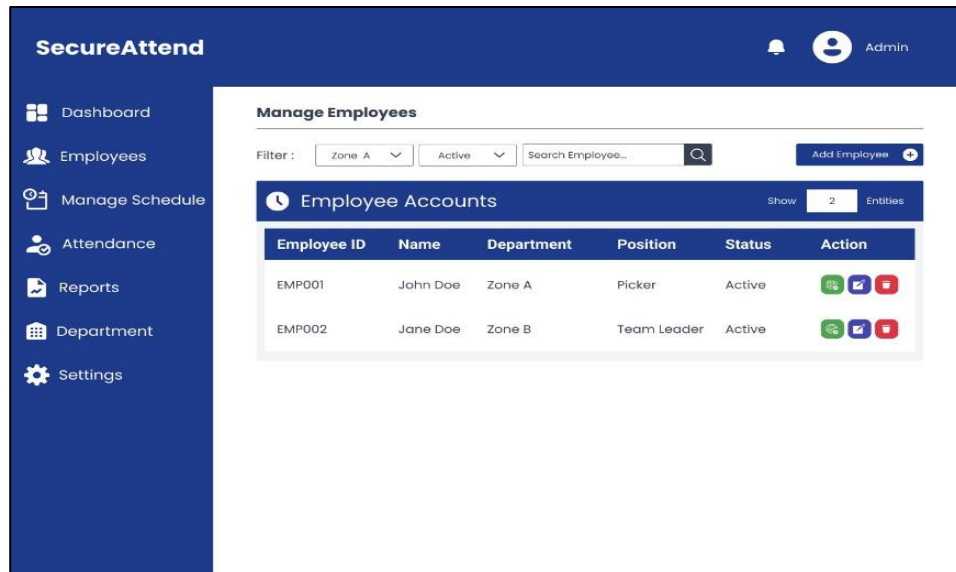


Figure 7. Wireframe of SecureAttend (Manage Employees)

This figure shows the Employee Management page. Admins can view a list of all employees with their department, status, and available actions. Buttons are provided for enrolling fingerprints, editing employee details, or deactivating employees. A search bar and filters help quickly locate specific employees, while an add employee button allows admins to register new staff.

Figure 8 shows the Manage Schedule page of the system.

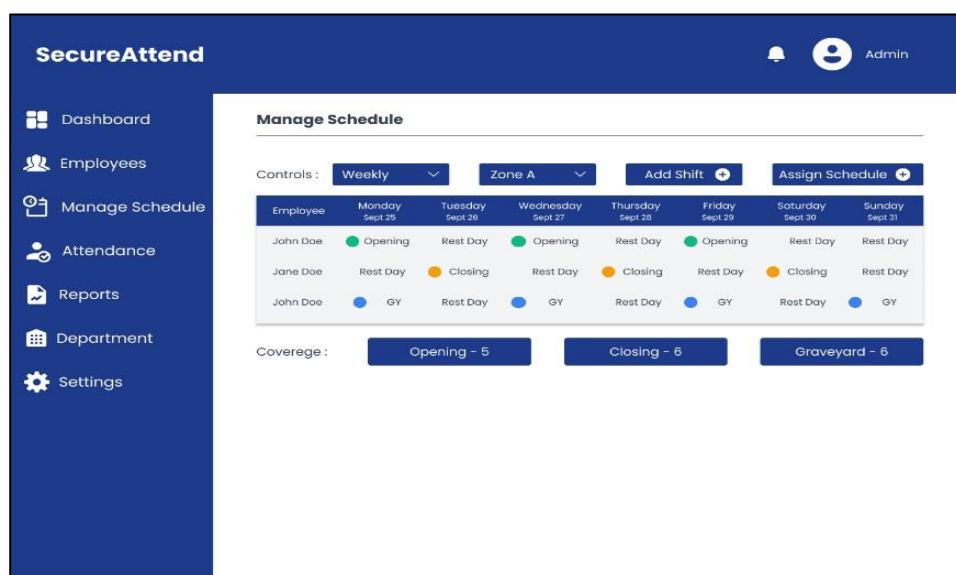
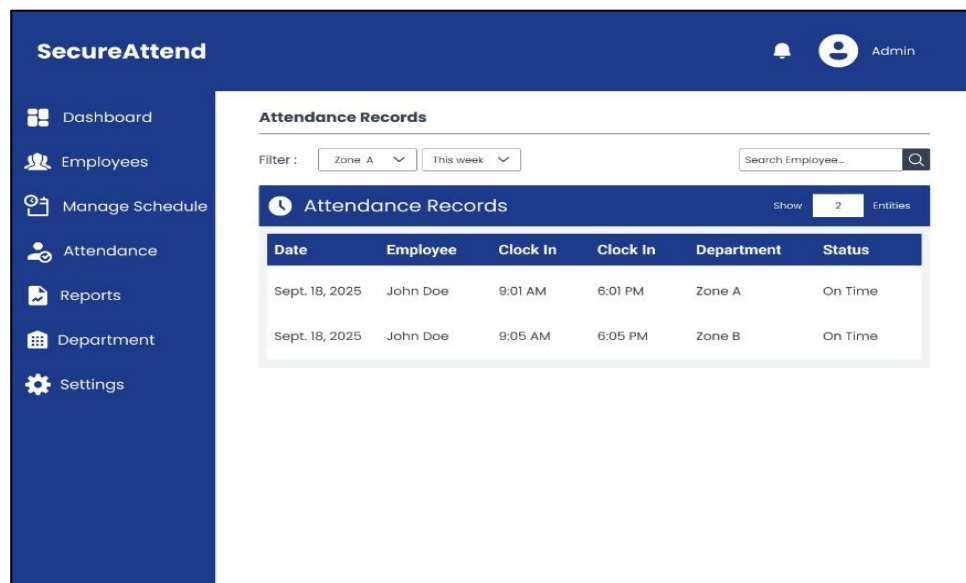


Figure 8. Wireframe of SecureAttend (Manage Schedule)

This figure shows the Schedule Management page. Admins can create and assign shifts to employees or departments using predefined templates such as morning, afternoon, or night shifts. The schedule is displayed in a grid or table format where each employee's daily assignment is visible. Coverage summaries and conflict alerts are included to prevent overlapping schedules and excessive workloads.

Figure 9 show the Attendance page of the system.



The image is a wireframe of the 'Attendance' page in a system called 'SecureAttend'. It features a dark blue sidebar on the left with navigation links: Dashboard, Employees, Manage Schedule, Attendance, Reports, Department, and Settings. The main content area has a header with the 'SecureAttend' logo, a notification bell, and a user profile labeled 'Admin'. Below the header, the 'Attendance Records' section includes a filter bar with 'Zone A' and 'This week' dropdowns, and a search bar labeled 'Search Employee...'. A table titled 'Attendance Records' shows two entries for 'John Doe' on 'Sept. 18, 2025', with clock-in and clock-out times, department (Zone A and Zone B), and status ('On Time').

Date	Employee	Clock In	Clock In	Department	Status
Sept. 18, 2025	John Doe	9:01 AM	6:01 PM	Zone A	On Time
Sept. 18, 2025	John Doe	9:05 AM	6:05 PM	Zone B	On Time

Figure 9. Wireframe of SecureAttend (Attendance)

This figure shows the Attendance Records page. It displays a table of attendance logs for all employees, including date, time in, time out, department, and status. Admins can filter logs by date range, department, or employee name. Export buttons are provided to download attendance records in Excel or PDF format for payroll and reporting.

Figure 10 shows the Reports page of the system.

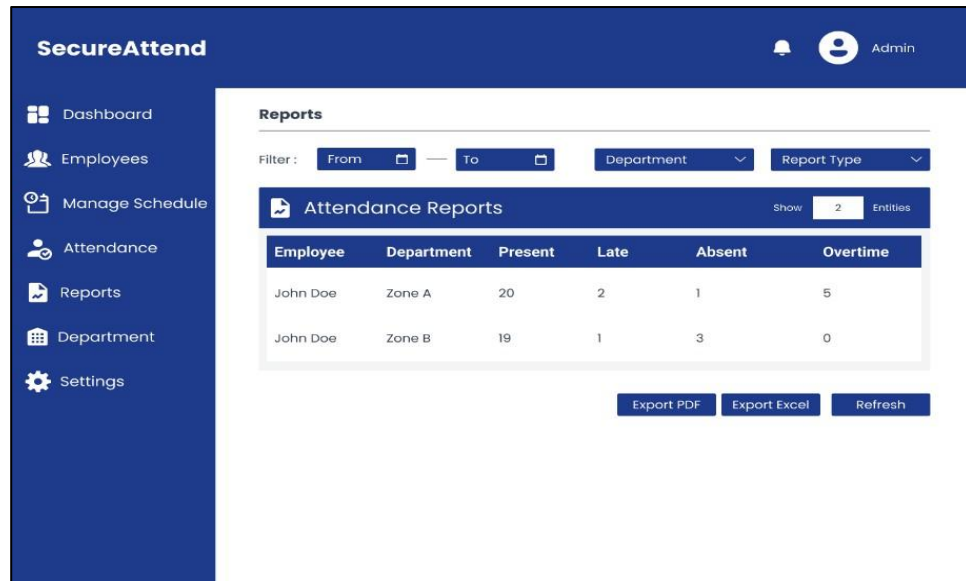


Figure 10. Wireframe of SecureAttend (Reports)

This figure shows the Reports page. Administrators can generate detailed reports for a specific date range or department. Summary cards highlight total presents, absents, lates, and overtime hours, visual charts for attendance trends, detailed employee table, and export option for reporting.

Figure 11 shows the Departments of the system.

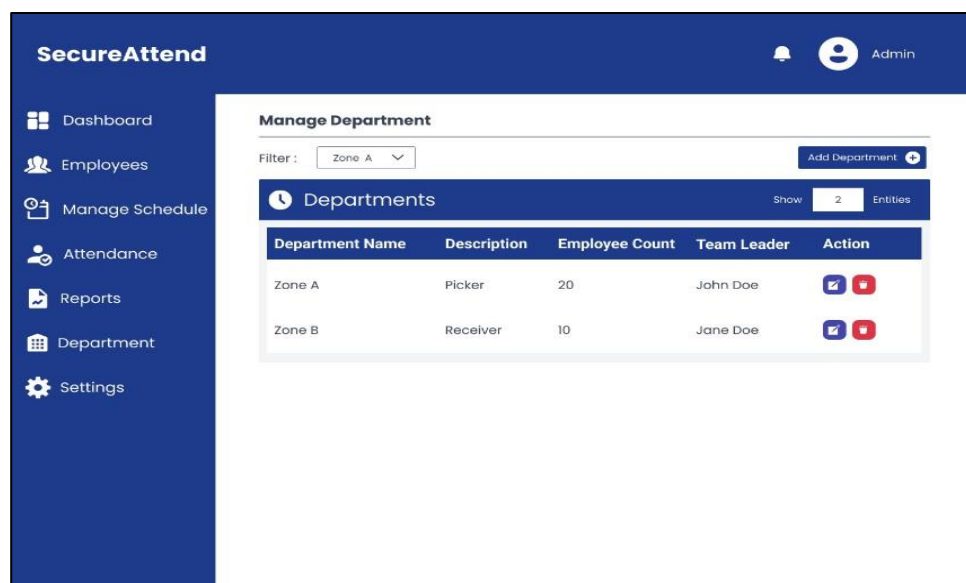


Figure 11 Wireframe of SecureAttend (Departments)

This figure shows the Manage Departments page for administrators. It lists all departments within the organization along with their assigned team leaders and the number of employees under each department. From this page, admins can add a new department, edit existing details, reassign a team leader, or deactivate departments that are no longer active. Action buttons are available beside each department for quick management, while a search bar and filters allow easy navigation across multiple departments.

Figure 12 shows the Biometric Attendance of the system.

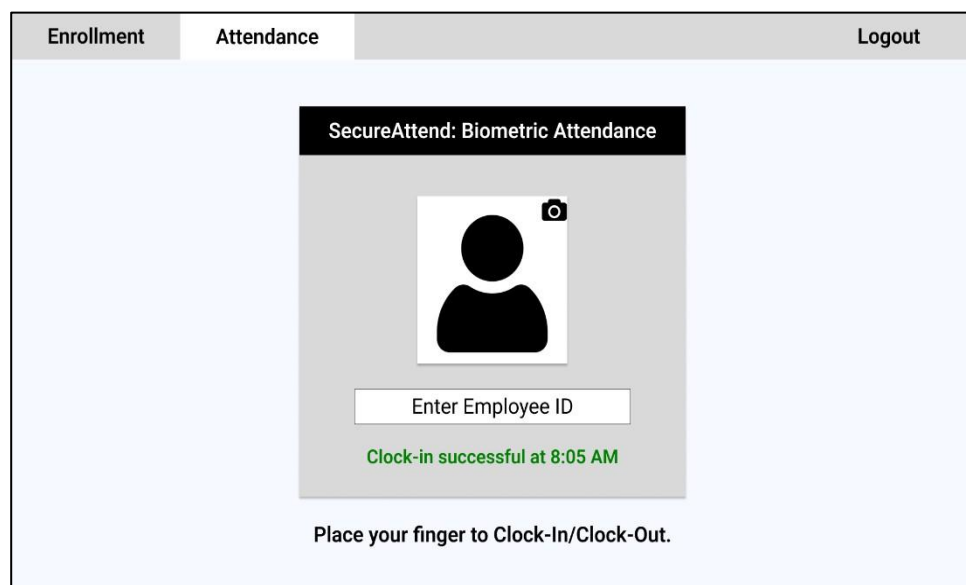
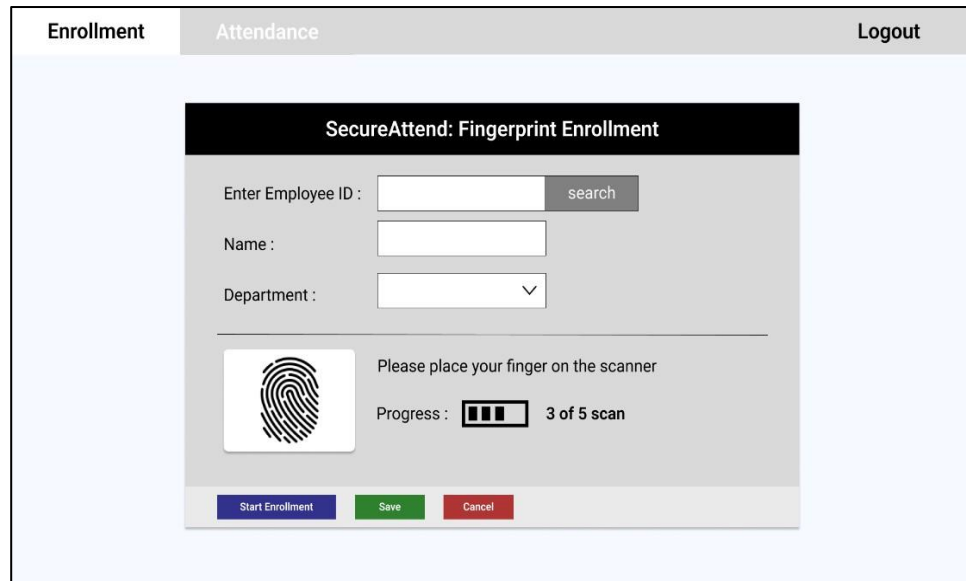


Figure 12. Wireframe of SecureAttend (Biometric Attendance)

This figure shows the Biometric Attendance interface, where employees clock in and clock out using the fingerprint scanner. The page displays the current date and time along with a simple prompt that instructs the employee to place their finger on the device. Once the fingerprint is recognized, the system validates the employee's schedule and confirms the attendance log with a success or failure message. This ensures that only authorized employees can record attendance through biometric verification.

Figure 13 shows the Biometric Enrollment of the system.



The wireframe shows a web application interface with a top navigation bar containing 'Enrollment', 'Attendance', and 'Logout' links. The main content area features a central modal titled 'SecureAttend: Fingerprint Enrollment'. This modal includes input fields for 'Enter Employee ID' (with a 'search' button), 'Name', and 'Department' (a dropdown menu). Below these fields is a fingerprint scanner icon and the instruction 'Please place your finger on the scanner'. A progress indicator shows '3 of 5 scan' with a bar chart. At the bottom of the modal are three buttons: 'Start Enrollment' (blue), 'Save' (green), and 'Cancel' (red).

Figure 13. Wireframe of SecureAttend (Biometric Enrollment)

This figure shows the Biometric Enrollment pop-up for administrators when registering a new employee or updating an existing record. The interface guides the admin through the enrollment process by asking the employee to scan their finger multiple times until the fingerprint template is successfully captured and saved. A progress indicator shows the status of the enrollment, and once completed, the fingerprint is securely linked to the employee's profile in the system. This step prevents fraudulent entries and ensures reliable identification during attendance logging.

Figure 14 shows the Entity-Relationship Diagram of the system.

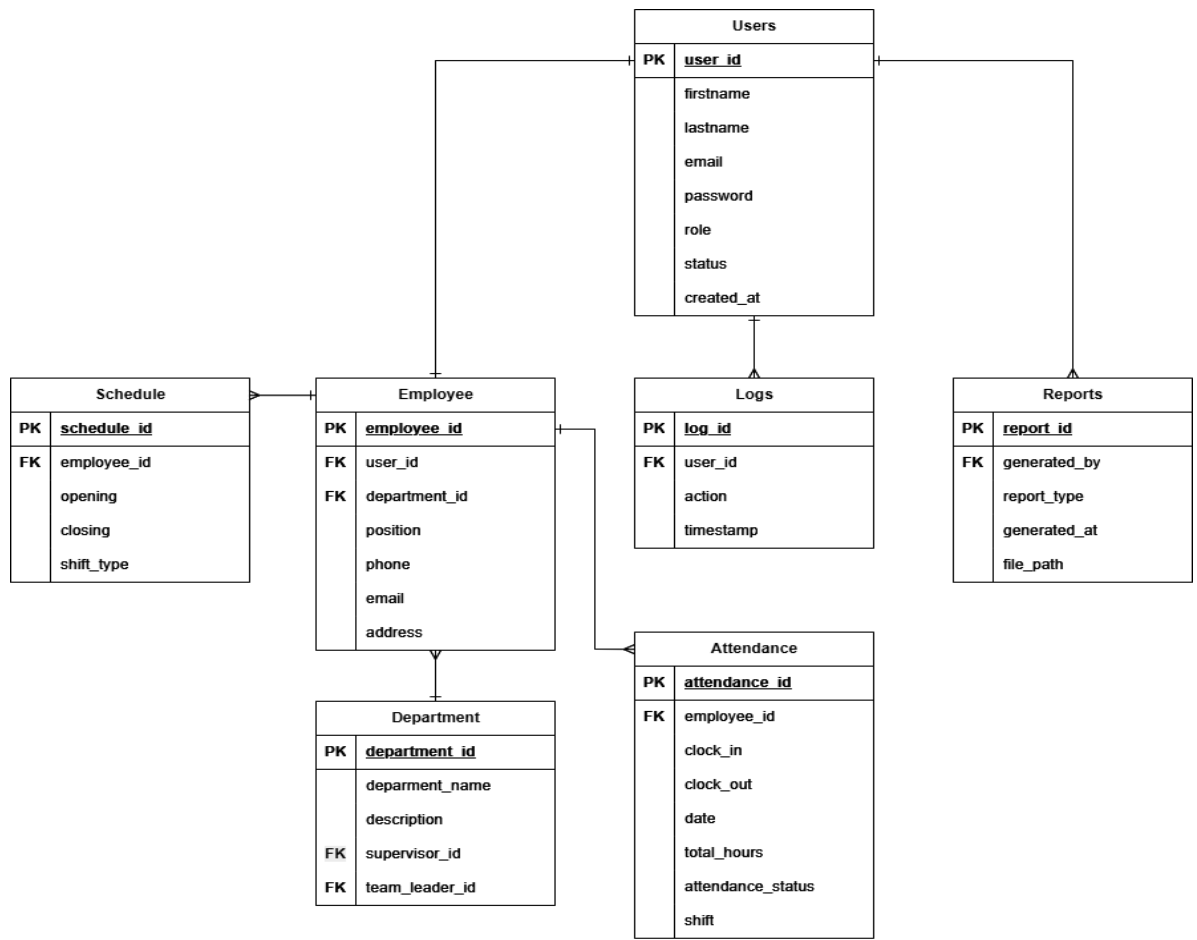


Figure 14. Entity-Relationship Diagram of Web-based Secureattend

The Entity Relationship Diagram (ERD) for the web-based SecureAttend illustrated how the core components of the system were structured and interconnected. At the center of the system was the Users table, which stored account credentials and role assignments for employees, supervisors, team leaders, and administrators. Each user was linked to the Employee table, which contained essential employee information such as department assignment, position, and contact details.

Employees were directly associated with the Attendance table, where biometric clock-in and clock-out records were stored along with attendance dates, total hours worked, assigned shifts, and attendance status. The Schedule table defined employee work schedules by recording shift times and shift types, ensuring proper workforce organization and attendance validation.

The Department table maintained the organizational structure by storing department information and assigned supervisors and team leaders, allowing attendance and schedule monitoring at the department level. The Reports table enabled authorized users to generate attendance summaries for payroll and monitoring purposes, while the Logs table recorded system activities to support auditing and accountability.

Together, these interconnected tables provided a centralized data structure for managing employees, schedules, departments, attendance records, and reports. This design supported secure role-based access and efficient workforce management within the SecureAttend system.

Figure 15. Entity-Relationship Diagram of Attendance Application Secureattend.

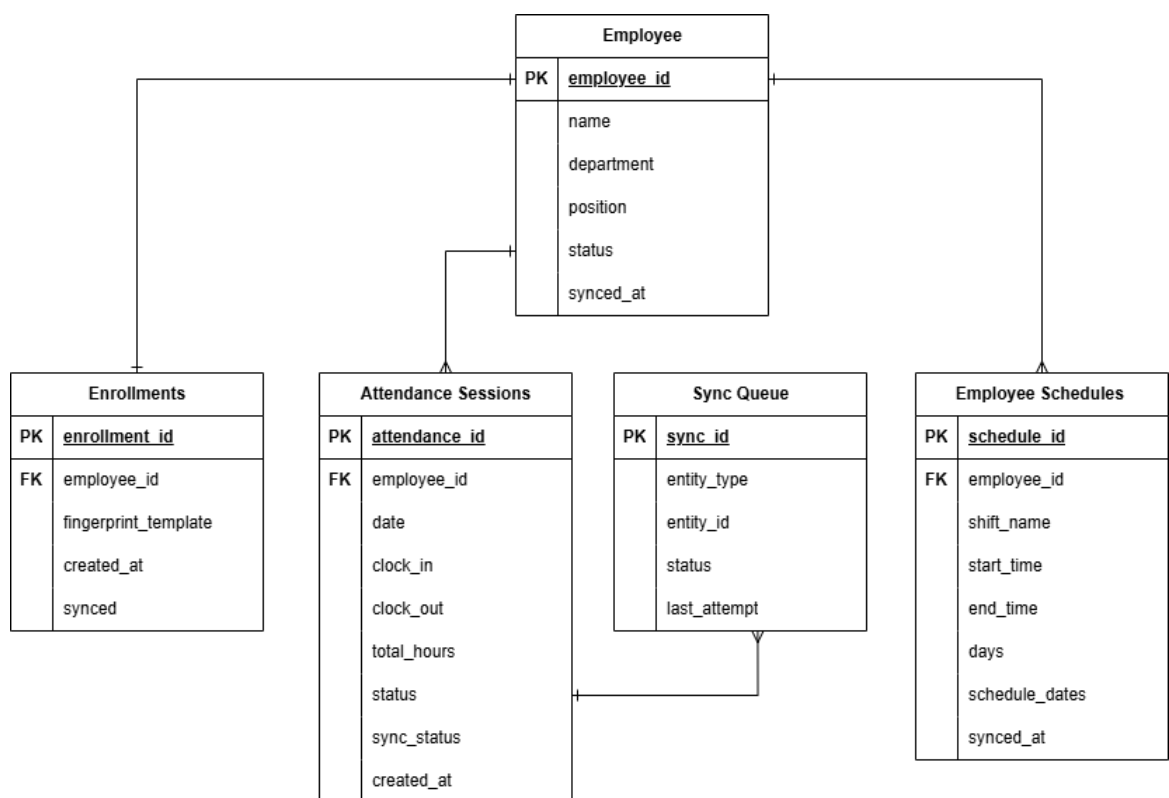


Figure 15. Entity-Relationship Diagram of Attendance Application Secureattend

The Entity Relationship Diagram (ERD) for the biometric standalone attendance application illustrated how attendance data was captured, stored, and synchronized independently of the web application. The system was designed to operate in both offline and online modes, with its database focused on biometric verification and attendance recording. At the core of the standalone system was the Employee table, which stored locally cached employee information synchronized from the web application and was used solely for identity reference during attendance verification.

The Enrollments table stored fingerprint templates associated with each employee, enabling biometric authentication during clock-in and clock-out activities. Attendance records were stored in the Attendance Sessions table, which logged employee clock-in and clock-out times, attendance dates, total hours worked, and attendance status. These records were saved locally to allow continuous operation even when network connectivity was unavailable.

The Employee Schedules table contained locally cached schedule information downloaded from the web application and was used to validate attendance based on assigned shifts. To support reliable data transmission, the Sync Queue table managed pending and completed synchronization tasks, ensuring that attendance and enrollment data were transmitted to the web application once connectivity was restored.

Together, these interconnected tables formed a lightweight and reliable database structure that enabled offline biometric attendance capture while maintaining seamless integration with the centralized web application. This design ensured data integrity, reliable synchronization, and uninterrupted attendance monitoring across different operational environments.

Figure 16. is the use-case diagram attendance management module.

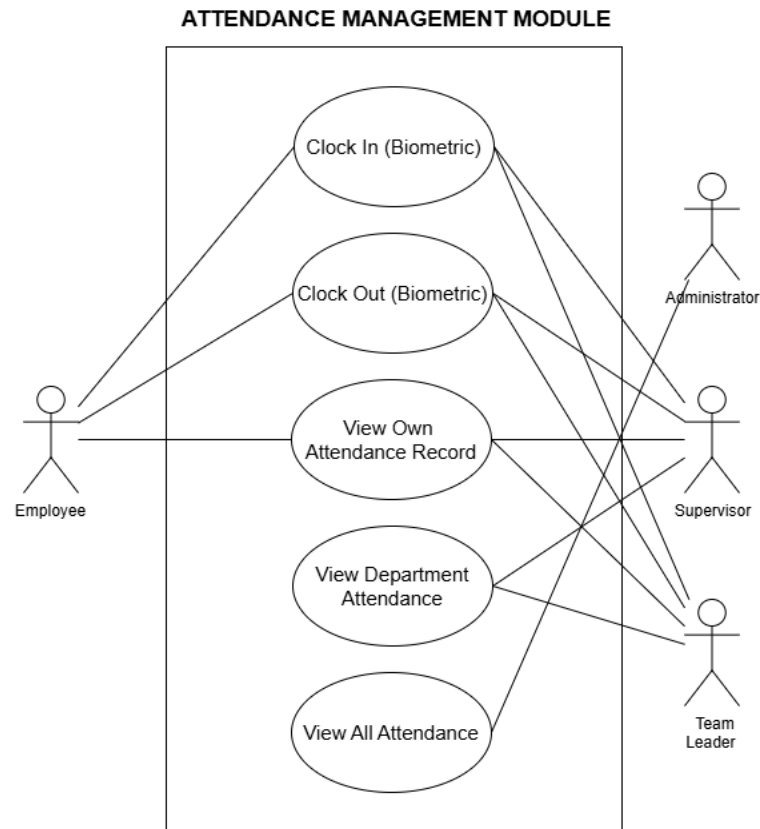


Figure 16. Use-case diagram for attendance management module of SecureAttend.

Figure 16 illustrated the use-case diagram for the attendance management module. The diagram showed how employee attendance was recorded and monitored using fingerprint biometric technology. Employees were able to clock in and clock out through the biometric device, while the system validated their identity and recorded attendance details such as time in, time out, and attendance status. Supervisors and team leaders were able to view and monitor attendance records of employees within their assigned departments, while administrators had access to company-wide attendance data for oversight and verification purposes.

Figure 17. is the use-case diagram employee management module.

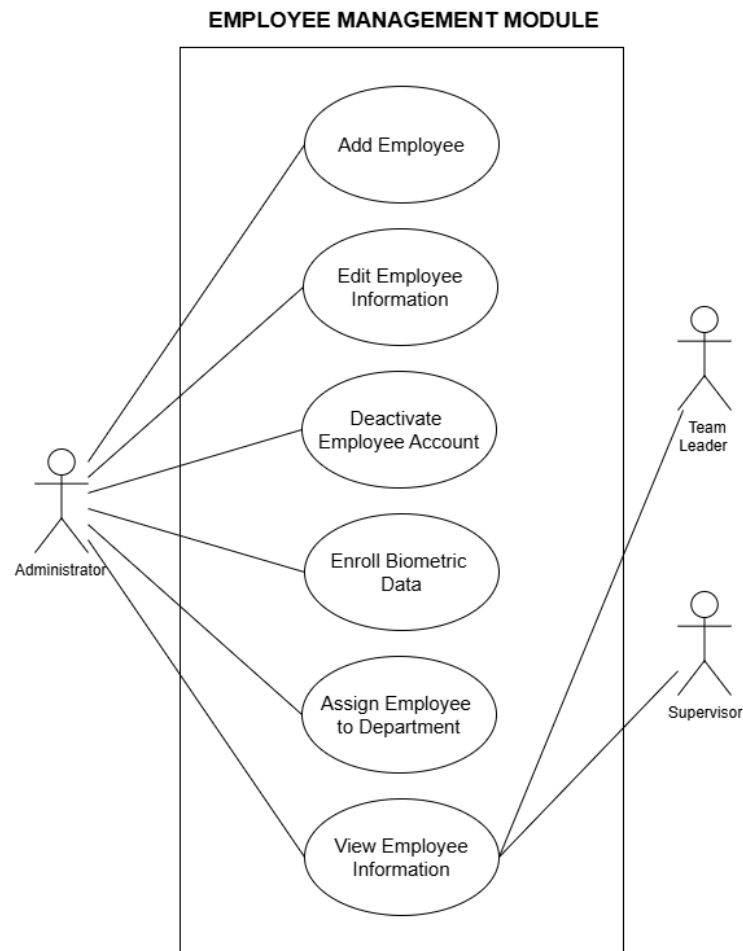


Figure 17. Use-case diagram for employee management module of SecureAttend

Figure 17 illustrated the use-case diagram for the user and employee management module. The diagram presented how employee accounts and user roles were managed within the system. Administrators were responsible for adding, updating, and deactivating employee profiles, assigning roles, enrolling biometric data, and assigning employees to departments. Supervisors and team leaders were allowed to view employee information within their respective departments for monitoring and coordination, ensuring proper role-based access control and secure system usage.

Figure 18. is the use-case diagram department management module.

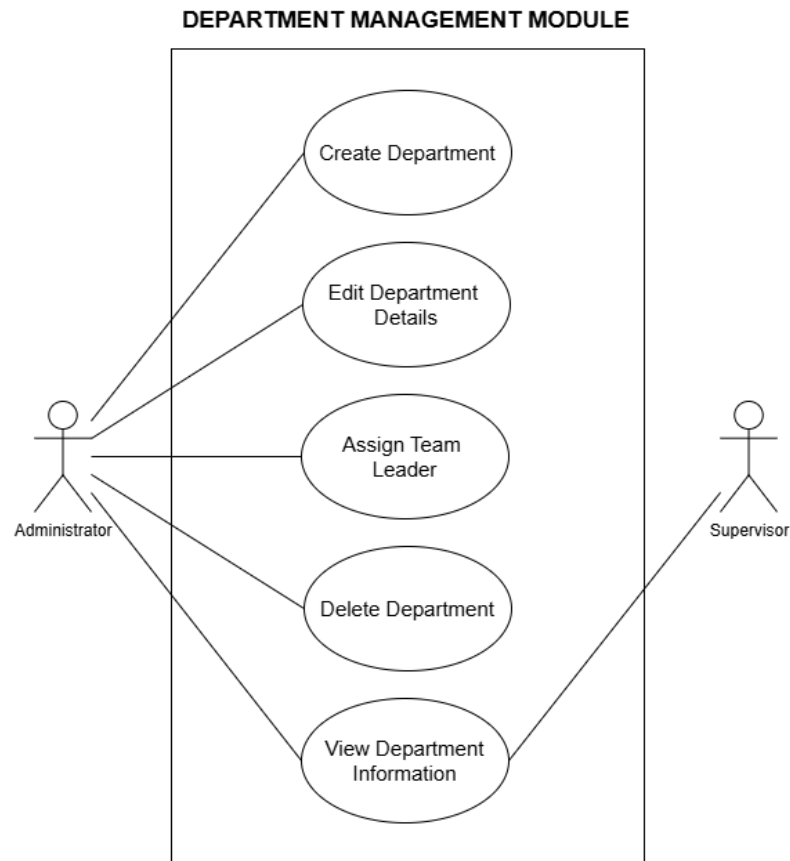


Figure 18. Use-case diagram for department management module of SecureAttend

Figure 18 illustrated the use-case diagram for the department management module of SecureAttend. The diagram showed how organizational departments were structured and maintained within the system. Administrators were able to create, update, and deactivate departments, as well as assign supervisors and team leaders. Supervisors were able to view department information and monitor attendance and schedules within their assigned departments. This module supported organized workforce management and effective supervision.

Figure 19. is the use-case diagram schedule management module.

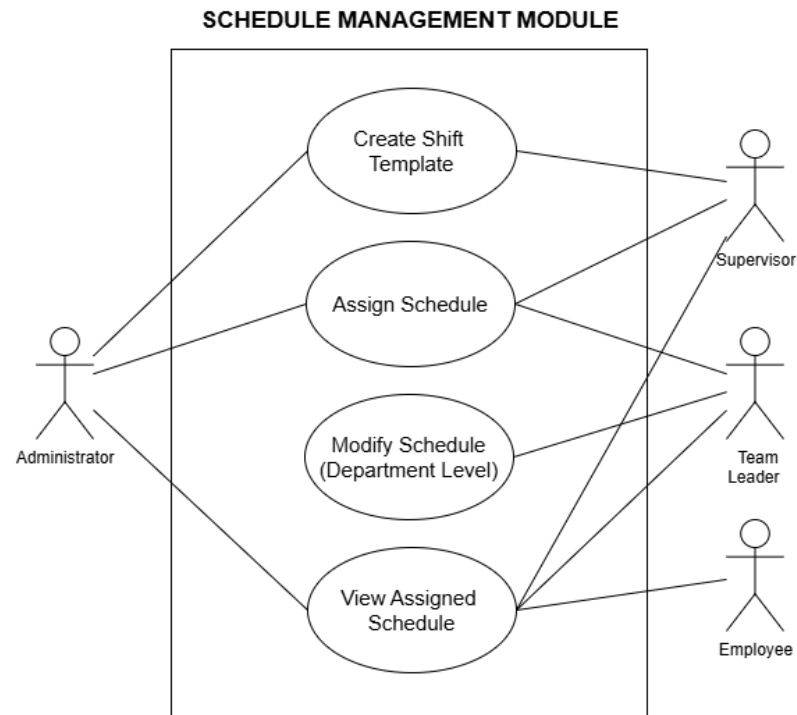


Figure 19. Use-case diagram for schedule management module of SecureAttend

Figure 19 illustrated the use-case diagram for the schedule management module. The diagram demonstrated how employee work schedules were created, assigned, and managed. Administrators were able to create shift templates and assign schedules, while supervisors and team leaders were able to modify schedules for employees within their departments to address operational needs. Employees were allowed to view their assigned schedules through the system. This module ensured proper workforce allocation and prevented scheduling conflicts.

Figure 20. is the use-case diagram reports module.

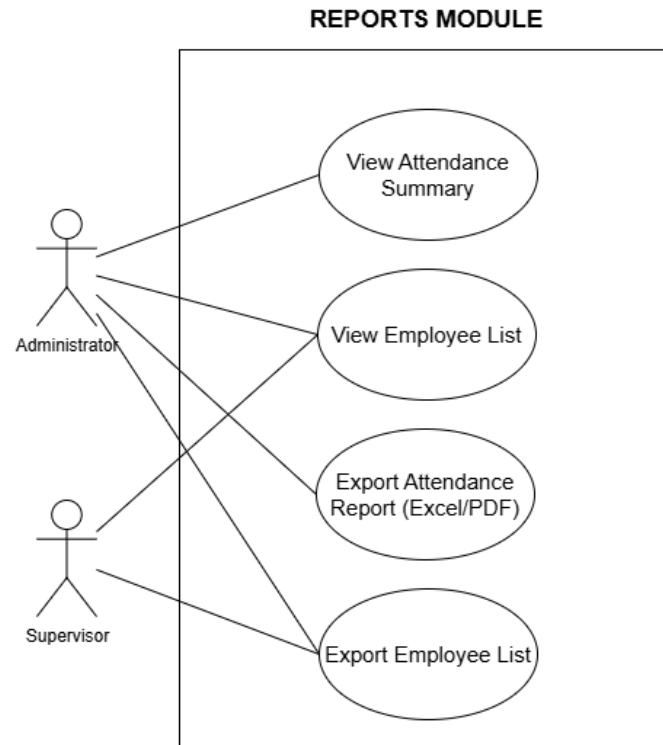


Figure 20. Use-case diagram for reports configuration module of SecureAttend

Figure 20 illustrated the use-case diagram for the reports module of SecureAttend. The diagram showed how attendance data was compiled into structured reports for monitoring and documentation. Administrators were able to view attendance summaries, employee lists, and export attendance reports in Excel or PDF format, as well as export employee lists for record-keeping and payroll purposes. Supervisors were also able to view attendance summaries and employee lists, and generate reports limited to their assigned departments. This module supported informed decision-making, accurate record management, and efficient reporting of workforce attendance.

Figure 21. is the use-case diagram system configuration module.

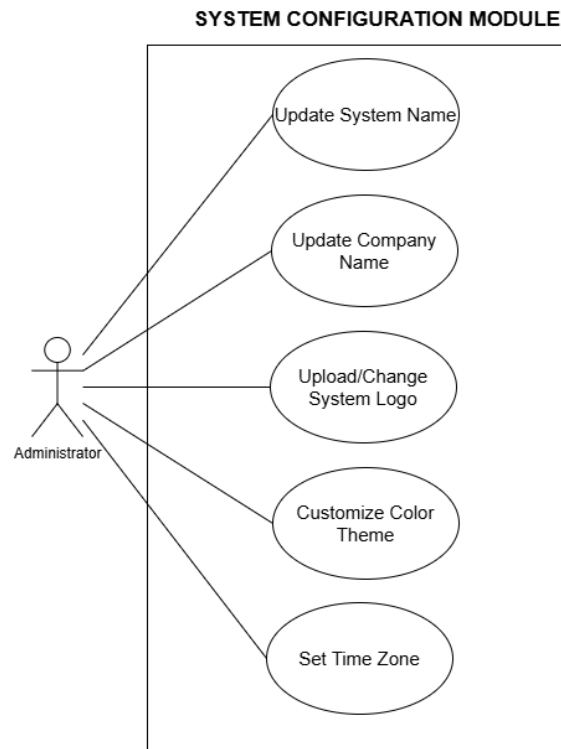


Figure 21. Use-case diagram for system configuration module of SecureAttend

Figure 21 illustrated the use-case diagram for the system configuration module of SecureAttend. The diagram showed how system-wide settings and configurations were managed to maintain system functionality and identity. Administrators were able to update system information such as the system name and company name, upload or change the system logo, customize color themes, and set the appropriate time zone. This module also supported the management of system appearance and operational settings to ensure consistency and alignment with organizational requirements.

Figure 22 illustrated the use-case diagram for the dashboard and notification module of SecureAttend. The diagram presented how system information and alerts were delivered to users. Employees were able to view attendance summaries and schedules through their dashboards. Supervisors and team leaders were able to monitor departmental attendance and receive alerts related to absences or schedule changes. Administrators had access to a comprehensive dashboard displaying company-wide

data. Notifications were sent to ensure timely communication and improved coordination among users.

Figure 22. is the use-case dashboard and notification module.

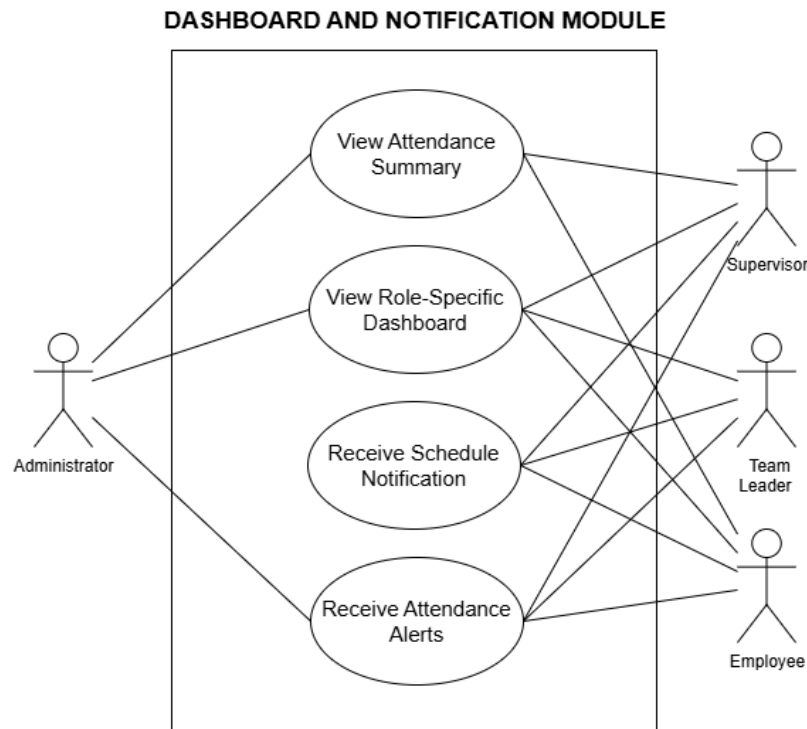


Figure 22. Use-case diagram for dashboard and notification module of SecureAttend.

Overall, the use case diagrams of SecureAttend, as shown in Figures 16 to 21, illustrated how the primary system actors Employee, Supervisor, Team Leader, and Administrator interacted with the system. Employees were provided with a convenient and secure way to record their attendance and view their schedules, while supervisors and team leaders were supported in monitoring attendance, validating records, and managing departmental operations. Administrators were facilitated in handling employee records, system configuration, and report generation to ensure accurate and efficient system management.

By integrating all attendance related functions into a single platform, SecureAttend went beyond automating attendance procedures; it improved accuracy,

enhanced operational transparency, and supported more organized and efficient workforce management.

Figure 23 shows the current business process of Toplis Solutions Inc.

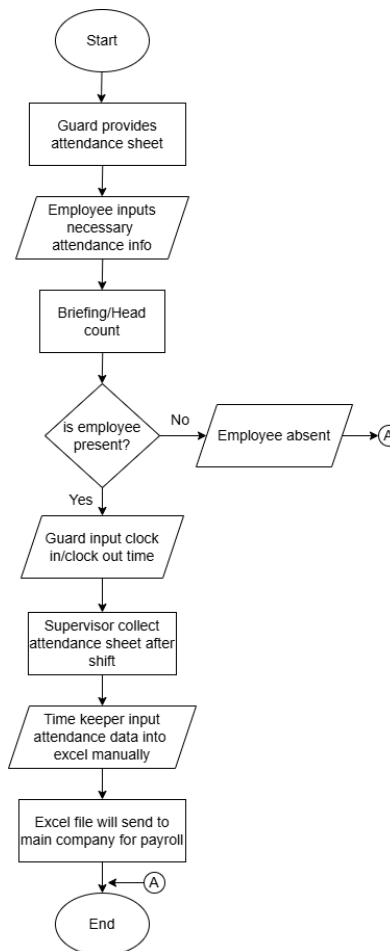


Figure 23 Business process flowchart

As indicated in Figure 23, the current attendance monitoring process of Toplis Solutions Inc. relied heavily on manual procedures that involved paper-based attendance sheets and the manual encoding of records. The process began when the guard distributed attendance sheets to employees at the start of the shift. Employees were then required to manually input their attendance information, after which a briefing and headcount were conducted to verify employee presence. During the shift, the guard manually validated clock in and clock out times through signage or handwritten confirmation.

After the shift ended, the attendance sheets were collected by the supervisor and forwarded to the timekeeper, who manually encoded the attendance data into an Excel file. The encoded file was then sent to the main company for payroll processing. This workflow was time consuming and prone to human error, as it depended on repetitive manual tasks and physical documents. The reliance on manual validation and data encoding often resulted in delays, inconsistencies in attendance records, and difficulty in maintaining accurate and timely attendance information.

Figure 24 illustrates the flowchart of SecureAttend.

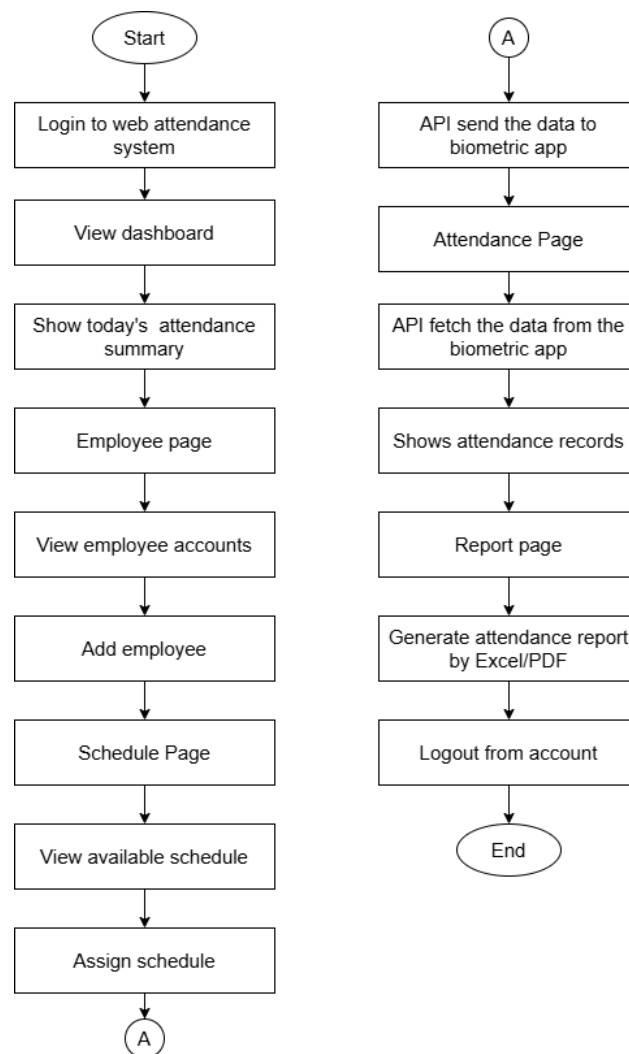


Figure 24. Secureattend Web-based Flow Chart Diagram

Figure 24 illustrated how the SecureAttend web-based attendance management system operated. The process began when the administrator logged into the web attendance system and was directed to the dashboard, where the current attendance summary was displayed. From there, the administrator proceeded to the employee page to view employee accounts and add new employees. The flow then continued to the schedule page, where available schedules were viewed and assigned to employees.

Through the system's API integration, data was sent to the biometric attendance application and later retrieved into the web-based platform for attendance monitoring. The administrator then accessed the attendance page to view attendance records and proceeded to the report page to generate attendance reports in Excel or PDF format. After completing these tasks, the user logged out of the account, which ended the process.

Figure 25 shows the continuation of SecureAttend's flowchart.

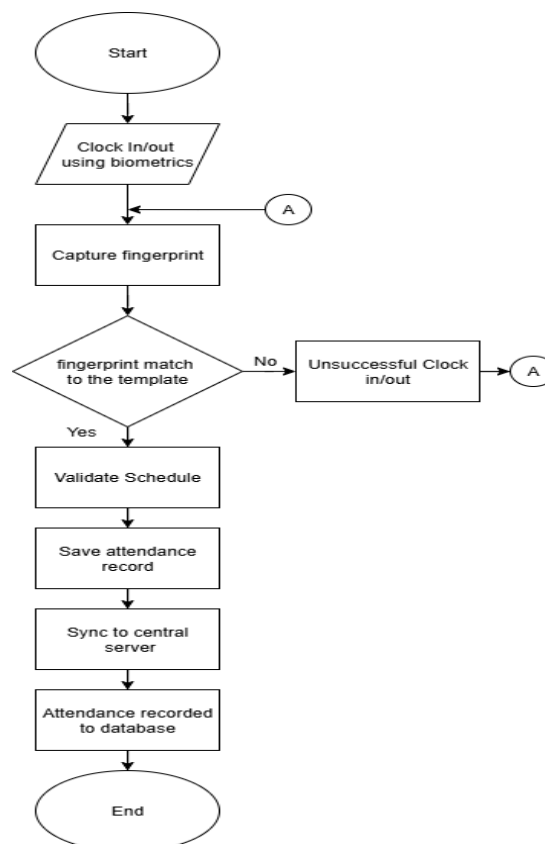


Figure 25. Secureattend Attendance App Flow Chart Diagram

Figure 25 illustrated how the biometric attendance application operated from login to attendance recording. The process began when users accessed the web based system by entering their credentials, after which the system directed them to role specific dashboards. Employees were able to view their schedules and attendance records, team leaders were able to manage team schedules and monitor attendance, while administrators had full control over employee management, schedule creation, department management, and report generation.

On the attendance application side, employees clocked in or out by scanning their fingerprints. The device validated the fingerprint against stored templates and checked the assigned schedule before saving the record. Attendance logs were stored locally on the device and were automatically synchronized to the central server once online, where they became accessible through dashboards and reports. This integrated approach ensured that employees recorded attendance through verified biometrics, team leaders managed their teams effectively, and administrators-maintained oversight with accurate data for monitoring and payroll purposes.

Requirement Analysis. SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite aimed to improve the manual attendance process of the organization. The primary users of SecureAttend were employees, team leaders, and administrators. The system provided the organization with a reliable, accurate, and secure attendance solution that improved the efficiency of the attendance process. To make the system more accessible to users, it was developed as a web-based platform. The existing attendance process of Toplis Solutions Inc. was performed manually; this process was prone to human error and time theft, which were the key problems that the proposed system addressed.

Operational Feasibility. SecureAttend had a number of potential challenges that could affect the operational feasibility of the proposed project.

Figure 26 showed a visual representation of these possible challenges. To

ensure the operational feasibility of the proposed project, these issues needed to be addressed.

Figure 26 shows the Fishbone Diagram of the system.

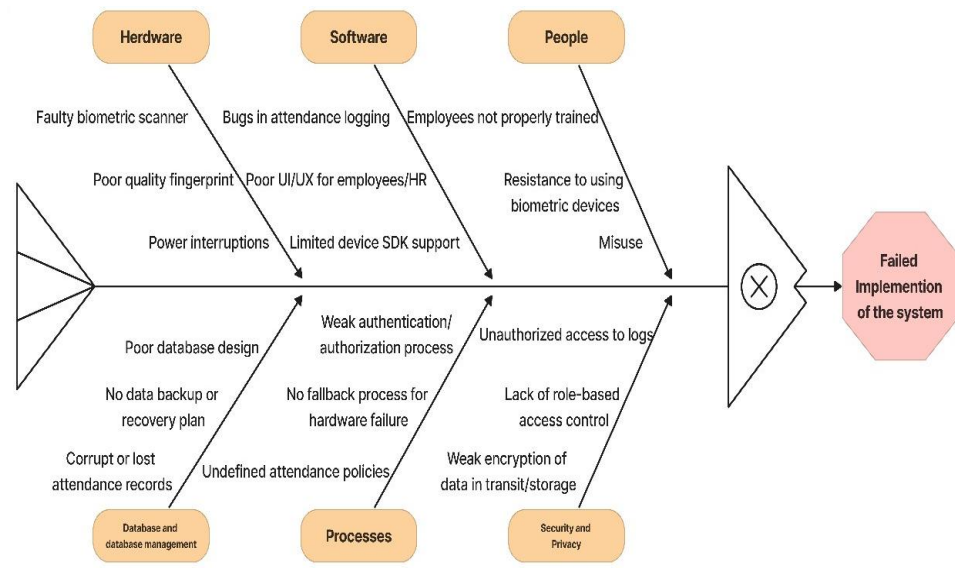


Figure 26. Fishbone diagram for the operational feasibility challenges in the development of SecureAttend

Technical feasibility. SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite was technically viable, as the necessary hardware, software, and technical expertise were available to support the development of the system. The system was developed using React.js, Node.js, Express.js, MySQL, and C#, which were sufficient to meet the project requirements. For deployment, the researchers utilized web hosting services such as Namecheap, Hostinger, and GoDaddy, which were suitable for organizations with multiple users and real time attendance tracking, while ZKTeco scanners were identified as appropriate for biometric attendance validation. The researchers possessed the necessary skills and technical knowledge required to execute and develop the proposed project and were capable of acquiring additional knowledge when needed. However, potential concerns related to scalability, data security, and user adaptability were identified

and addressed through the implementation of role-based access control, optimization of backend and API performance, and provision of proper user training. With these measures in place, the proposed project was executed with effective risk management and optimized strategies, ensuring efficiency and reliability. (see Appendix 1).

Economic feasibility. The operational costs required for the implementation of the proposed system were determined based on its economic feasibility evaluation (see Appendix 2). The cost breakdown was simplified to ensure affordability while maintaining system functionality. The primary expenses included the biometric module amounting to ₱2,000, which was used for attendance recording, web hosting services costing ₱2,500 to support the system's online accessibility and performance, and miscellaneous expenses estimated at ₱2,000 to cover additional operational needs. These costs resulted in a total initial expenditure of ₱6,500, indicating that the system was financially viable due to its low implementation cost and minimal operational expenses. Overall, these expenditures demonstrated the cost efficiency and sustainability of the project, allowing effective system operation without imposing a significant financial burden on the organization.

Schedule feasibility. SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite was conducted from February 2025 to January 2026, as presented in the Gantt chart (see Appendix 3). The project began with proposal writing from February to March 2025, during which the scope, objectives, and methodology were defined. This was followed by the proposal defense in April 2025, which confirmed the feasibility of the study prior to development. System design was carried out in April 2025, focusing on system architecture, database structure, and user interface design. System development was conducted from March 2025 to September 2025 and included coding, database integration, and API implementation.

Following development, system testing was performed in September 2025 to evaluate functionality, security, and efficiency. The system evaluation phase, also conducted in September 2025, assessed the performance of the project based on user acceptance and project requirements. In October 2025, system implementation was carried out, during which the system was formally deployed at Toplis Solutions Inc., Maguyam, Silang, Cavite. Manuscript preparation was conducted from October 2025 to December 2025, followed by the final oral defense in December 2025. Finally, manuscript review was completed in January 2026 to ensure that the final document was properly refined before submission. This structured timeline ensured systematic development, testing, and evaluation, leading to the effective implementation of the attendance monitoring system.

Requirement Documentation. SecureAttend was a web connected attendance management system designed to address the attendance problems of Toplis Solutions Inc., Maguyam, Silang, Cavite by improving the attendance process through biometric technology. The system aimed to manage employee attendance and track the number of employees who had clocked in, thereby preventing overcrowding. It utilized biometric technology to prevent buddy punching and ensure accurate identity verification. Additionally, the system allowed administrators to export generated attendance reports into Excel format, which were then submitted to the main company of Toplis Solutions Inc. for payroll processing.

User and employee management module. This module manages user accounts and employee records within the system. Administrators can add, update, and deactivate employee profiles, assign roles, and enroll fingerprint biometric data. It also allows the assignment of employees to specific departments and ensures that role-based access is properly enforced. By centralizing employee information and access control, the module supports secure system usage and efficient workforce management.

Attendance management module. This module handles the recording and validation of employee attendance using fingerprint biometric technology. Employees can clock in and clock out through the biometric device, while the system verifies identity and checks assigned schedules before logging attendance. Attendance records include time in, time out, total work hours, and attendance status. This module helps prevent attendance fraud, such as buddy punching, and ensures accurate and reliable attendance tracking.

Schedule management module. This module allows administrators and team leaders to create, assign, and modify employee work schedules. It supports the use of predefined shift templates and validates schedules to prevent conflicts and overlapping assignments. Employees can view their assigned schedules through the system, while team leaders can manage schedules for their respective teams. This module helps ensure proper workforce allocation and operational efficiency.

Department management module. This module enables administrators to organize employees into departments or zones. It allows the creation, updating, and deactivation of departments, as well as the assignment of team leaders. By structuring employees according to departments, the module supports effective supervision, attendance monitoring, and schedule management within the organization.

Notification and alert module. This module automates system communication by sending notifications and alerts to employees and team leaders. Notifications may include schedule updates, attendance confirmations, and important system announcements. By delivering timely information, the module improves coordination, reduces miscommunication, and ensures that users remain informed of relevant system activities.

Reports module. This module enables administrators to generate structured attendance reports that support monitoring and documentation of

employee attendance. It compiles attendance data such as employee presence, lateness, and absences, which can be filtered by date range, department, or individual employee. The generated reports can also be exported for payroll processing and record-keeping purposes. By providing organized and accessible attendance information, this module helps administrators monitor workforce attendance and supports informed decision-making within the organization.

System Configuration Module. This module allows administrators to manage important system settings necessary for the proper operation of the SecureAttend system. It enables administrators to modify system information such as the system logo, company name, system name, color theme, and time zone. These configuration options allow the system to align with the organization's identity and operational requirements while maintaining consistent system appearance and functionality.

System Development

The development of the SecureAttend system utilized the Modified Waterfall Model, which allowed a structured yet flexible approach to system development. Unlike the traditional linear Waterfall model, the modified version enabled feedback and revisiting of earlier phases when issues were identified, thereby reducing the risk of major errors later in the project. Each stage, including requirements, design, implementation, testing, and maintenance, remained clearly defined and documented, while the model's iterative nature ensured continuous refinement (Nugroho & Izzah, 2018). This approach balanced structure with adaptability, making it suitable for the development of SecureAttend.

Figure 27 shows the chosen SDLC of the proponents.

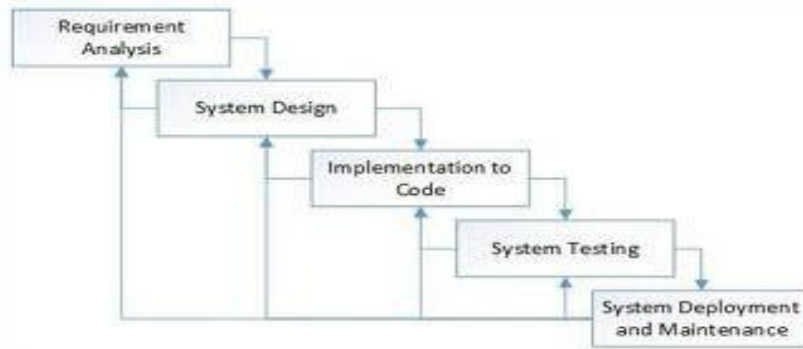


Figure 27. Modified Waterfall Method (Nugroho & Izzah, 2018)

Requirement Analysis. This phase focused on gathering information to define the system's requirements. Data was collected through direct interviews with a Toplis Solutions Inc. employee to understand existing attendance procedures. The system supported three types of users: administrators, who had access to all system functionalities including managing and monitoring employee attendance; team leaders, who managed and monitored attendance within their respective departments and members; and employees, who were able to view their attendance records and schedules.

System Design. In this phase, the system architecture, workflows, and interface design were developed using Unified Modelling Language (UML) diagrams, including use case diagrams and flowcharts. The database design was also developed through the use of an Entity Relationship Diagram (ERD), while a high-fidelity wireframe was created for the user interface.

Implementation. In this phase of the life cycle, the system was developed based on the design created in the previous phase. The system was built using the JavaScript programming language with the React 19.0 framework and C# for biometric integration.

System Testing. In this phase, the system was tested to ensure that it worked as intended. Various tests were conducted to validate the system's usability, accuracy, and reliability. The system was also tested in real world scenarios to ensure it performed

effectively in actual operational environments. System quality was assessed using ISO 25010 standards.

Operation and Maintenance. This phase involved the deployment of the system to the client. During this phase, the system's performance was monitored to ensure that it met expectations and operated smoothly in actual operational environments. This phase also included system maintenance, ensuring that the system remained functional and up to date.

System Testing

The system testing phase ensured that SecureAttend: An Attendance Management System Using Biometrics for Toplis Solutions Inc., Silang, Cavite met its functional and non functional requirements. The system was evaluated using the testing template (see Appendix 4) provided by the department to verify that all modules performed as intended. The testing process covered the test environment, functionality, portability, usability, and security.

The test environment ensured that the proposed system performed correctly in different scenarios that mimicked actual usage conditions. This included testing the biometric device integration for offline attendance logging and verifying that the web-based platform operated smoothly across different operating systems and screen resolutions. Functionality testing confirmed that all system modules operated as expected by checking whether employees successfully recorded attendance using the fingerprint scanner and whether attendance records were properly stored and synchronized. It also verified that employees viewed their schedules and attendance records, team leaders monitored and managed schedules for their members, and administrators managed employees, departments, schedules, and reports. Each feature was carefully tested to ensure accuracy, consistency, and reliability of system operations.

An IT expert played a crucial role in evaluating the system's overall architecture and ensuring that its design and implementation aligned with best practices in software

development. The expert assessed the system's technical structure, security protocols, and data management processes, and provided professional recommendations related to system optimization, scalability, and sustainability. This evaluation ensured that the SecureAttend system met technical standards and complied with industry practices for secure and efficient software systems.

Portability testing ensured that the system was accessible across various devices and platforms. As the system was web based, it was tested on different browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge to ensure compatibility. It was also accessed using mobile devices, including smartphones and tablets, to verify that the interface adapted properly to different screen sizes and remained user friendly across platforms.

Usability testing focused on evaluating the ease of use and clarity of the SecureAttend system interface, and recommended improvements were implemented to enhance navigation and user interaction. Security testing verified the effectiveness of authentication mechanisms and role-based access control to ensure that only authorized users accessed specific system functions. Feedback from testers was utilized to identify system issues, guide necessary corrections, and refine system features to improve reliability and performance. All responses were subjected to structured data treatment, where results were classified as *Complied*, indicating that system requirements were met, or *Not Complied*, indicating the need for improvement. Through this multi-level testing approach, the system was validated in terms of reliability, functionality, and adaptability, confirming its readiness for deployment and its capability to replace manual attendance procedures with an accurate, efficient, and reliable attendance management system.

System Evaluation

System evaluation was the stage where a fully developed system was evaluated in order to assess its overall performance, functionality, and usability of SecureAttend:

An Attendance Management System Using Biometrics for Toplis Solution Inc, Silang Cavite.

Product quality. It aimed at determining the quality of SecureAttend attendance system according to the ISO 25010 standard product quality guidelines. It collected perceptions from 5 IT specialists, 3 domain expert, and 72 employees selected from within the organization. Their evaluation helped determine if the SecureAttend attendance system was efficient, user friendly, and capable of simplifying attendance management and reporting processes. The evaluation form that was used was the DIIT Evaluation Form for ISO Product Quality (see Appendix 5). It used a five-point Likert scale which ranged from 1 (poor) to 5 (excellent). It measured the functionality, performance, usability, security, maintainability, and portability of the system. Functional suitability (indicators 1-3) evaluated how fitting and complete the system's features were in fulfilling its intended functions. Performance efficiency (indicators 4-6) assessed the system's capability to perform tasks effectively while improving time and resources. Compatibility (indicators 7-8) measured how well the system interacted and integrated with other systems within the same environment. Usability (indicators 9-15) examined how easy the system was to learn, use, and understand. Reliability (indicators 16-18) highlighted the consistency and stability of the system during operation. Security (indicators 19-22) focused on protecting the data and information stored within the system. Maintainability (indicators 24-28) evaluated how easily the system could be modified, updated, or repaired when necessary. Lastly, Portability (indicators 29-31) evaluated the system's capability to be transferred, installed, or operated across different hardware or software environments without difficulties. This input helped determine the strengths and weaknesses of the application. It confirmed if the application achieved its objectives, while end users were satisfied with the experience.

Quality-in-use. It evaluated the system effectiveness in accessing and utilizing the systems' features in a real-world scenario. The evaluation form for quality in use (see Appendix 6) also used a five-point Likert scale, with 1 being poor and 5 being excellent.

The administrator assessed the effectiveness of the system in a real workplace. The system administrator, who served as the primary user, was the main respondent during the system testing phase. The evaluation aimed to assess the system's quality, effectiveness, maintainability, and monitoring capabilities. Feedback and observations were gathered based on the administrator's direct interaction with the system throughout the evaluation process. Relevant system data and operational insights were also considered during the assessment. Effectiveness (indicator 32) measured how successful or unsuccessful the system was for users in accomplishing their goals. Efficiency (indicator 33) measured how many resources were used to achieve these goals. Satisfaction (indicators 34-37) checked how well the system met user expectations and needs when it was used in a real environment. Freedom from Risk (indicators 38-40) determined how the system performed in mitigating risk, including its capability in addressing threats ranging from environmental to health, safety, and economic risks. As a final check, Context Coverage (indicators 41-42) ensured that the system worked efficiently in the given settings. Thus, this thorough evaluation verified that the system catered to user needs while providing them with a reliable, functional, and consistent experience in different situations.

The proposed system for SecureAttend Attendance System used a simple random sampling technique and a survey questionnaire to gather relevant information needed for the study. Respondents were required to rate their experience with the system using a Likert scale from 1 to 5, where 1 meant poor and 5 meant excellent. The study applied the weighted mean formula $WM = \frac{\sum(fx)}{\sum f}$ to evaluate the survey results gathered from selected users. In this formula, "f" represented the frequency, while "x" represented the assigned rating. By applying this computation, the researchers calculated the average perception of the respondents. The mean was computed for each of the system criteria. The results were then interpreted using descriptive ratings such as very good, good, fair, and poor to determine if the system met the expected standards. Through this process, it was assessed whether SecureAttend: Attendance Management

System using Biometrics for Toplis Solutions Inc., Silang, effectively enhanced and streamlined the traditional manual attendance process.

Implementation Plan

The SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite was implemented in several stages to ensure a smooth transition. These stages included preparation, deployment, user training, and monitoring.

First was the preparation phase, during which the researchers procured the necessary equipment and software. This phase also included arrangements for web hosting and the installation of biometric devices.

The second stage was the deployment phase. During this phase, a beta version of the system was deployed to a selected group of staff members. The primary objective was to test whether the attendance system functioned as intended prior to full deployment.

The third stage was the user training phase. This phase involved training employees and supervisors on how the system operated. A system manual and training sessions were provided to help users understand and effectively use the system.

In the monitoring and evaluation phase, the researchers continuously assessed the system's performance based on user feedback. Performance indicators included availability, security, and user engagement. Regular updates and maintenance were conducted to improve system performance and resolve identified issues.

Key factors that were considered included user friendliness, security, safety, and maintainability of the system over time. SecureAttend was developed as a reliable and efficient solution for attendance management.

RESULT AND DISCUSSION

This chapter presents the organization's system, highlighting its structure capabilities, weaknesses, and expert evaluations.

System Design

The system design of SecureAttend for Toplis Solutions Inc. focuses on developing a secure and web-connected attendance management system using fingerprint biometrics to improve attendance accuracy, prevent buddy punching, and reduce manual attendance processing. It includes essential modules such as user and employee management, attendance, schedules, department, reports, a system configuration, and notification to support the organization's daily operations. The system also provides role-based access and an offline biometric application with data synchronization, ensuring accurate attendance recording, smooth navigation, and effective workforce management.

User and employee management module. This module allows the administrator to manage employee accounts and system users within SecureAttend.

EMPLOYEE ID	NAME	DEPARTMENT	POSITION	STATUS	ACTIONS
TSI00061	Clarenz Reyes	Zone E	Team Leader	Active	[Icons]
TSI67890	Wence Lontok	Company-wide	Warehouse Admin	Active	[Icons]
TSI00111	James Tojan	Zone A	Reach Truck Operator	Active	[Icons]
TSI12345	Lorence Rodriguez	Zone A	Helper	Active	[Icons]
TSI00001	Resty Eliana	Company-wide	Supervisor	Active	[Icons]
TSI00003	Fernando Dela Cruz	Zone A	Team Leader	Active	[Icons]
TSI00004	Kenny Siatita	Zone B	Team Leader	Active	[Icons]

Figure 28. Screenshot of the employees' page of the system

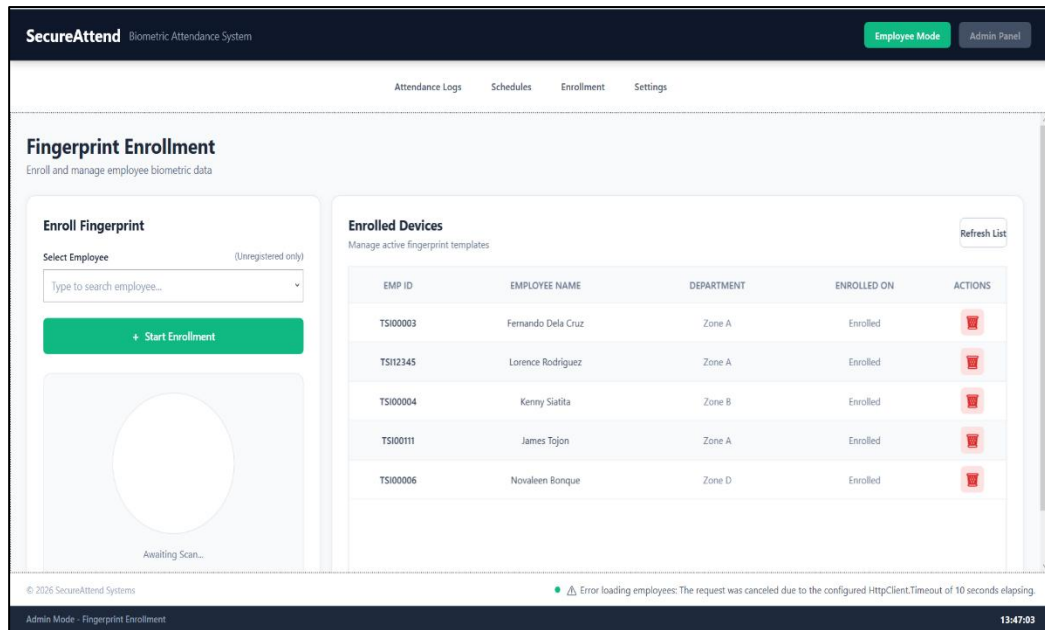


Figure 29. Screenshot of the fingerprint enrollment of the system

The module provides the functionality to view and manage all employee records within the system. Administrators can add new employees, update employee details, and assign positions and departments. The system also provides options for enrolling employee fingerprints in attendance application in figure 29, editing employee information, and activating or deactivating employee accounts when necessary. Through the use of search and filter options, administrators can easily locate employee records and manage large numbers of employees efficiently. This module ensures that employee information is organized, accurate, and securely stored within the system.

Attendance management module. This module allows employees to record their attendance using fingerprint biometric authentication.

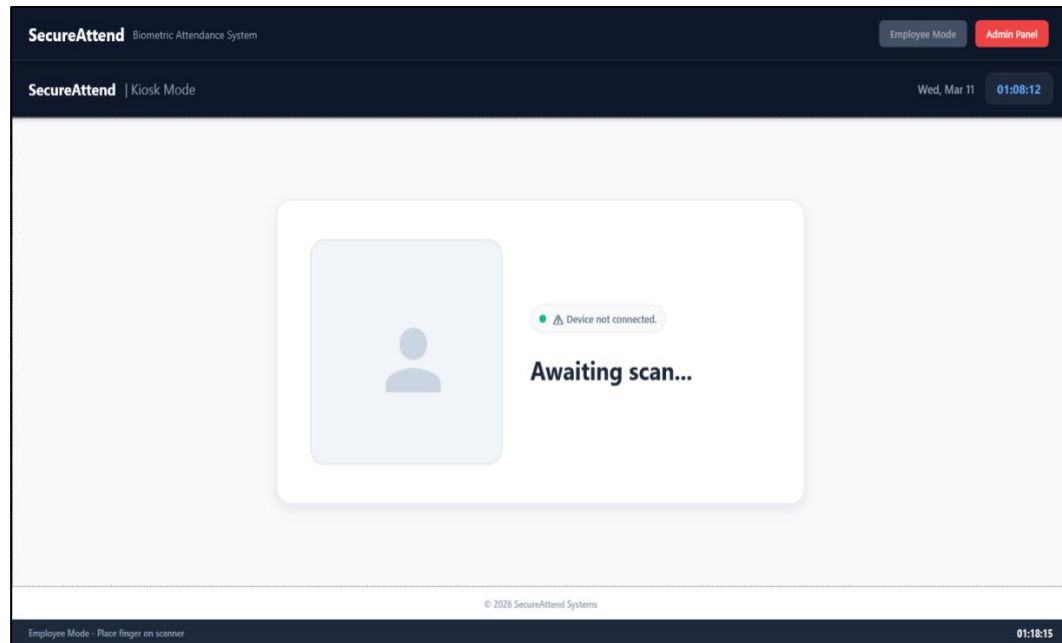


Figure 30. Screenshot of the attendance biometric app of the system

DATE	EMPLOYEE	CLOCK IN	CLOCK OUT	HOURS	DEPARTMENT	STATUS	ACTIONS
Mar 11, 2026	Fernando Dela Cruz	-	-	-	Zone A	Absent	-
Feb 28, 2026	Lorence Rodriguez	21:17:50	-	-	Zone A	Late	-
Feb 28, 2026	James Tojon	21:03:56	21:09:19	0.09 hrs	Zone A	Late	-
Feb 20, 2026	Kenny Siatita	-	-	-	Zone B	Absent	-
Feb 20, 2026	James Tojon	-	-	-	Zone A	Absent	-
Feb 20, 2026	Fernando Dela Cruz	09:43:08	-	-	Zone A	Missed Clock-out	-
Feb 20, 2026	Lorence Rodriguez	09:43:02	-	-	Zone A	Missed Clock-out	-

Figure 31. Screenshot of the attendance page of the system

Figures 30 and 31 present the attendance recording interface of the attendance application and the attendance records module of the web-based SecureAttend system. Figure 30 shows the attendance application where employees record their attendance using fingerprint biometric authentication. Employees perform clock-in and

clock-out by scanning their fingerprints through the biometric device connected to the attendance application. Once the fingerprint is recognized, the system verifies the employee's identity and records the attendance information, which is then transmitted to the central database. Figure 31 displays the web-based attendance records module where administrators can view and monitor the recorded attendance data. The interface presents a structured table showing details such as date, employee name, clock-in time, clock-out time, working hours, department, and attendance status. Because the attendance application and the web system are connected, the recorded attendance data is automatically synchronized, allowing administrators to track and manage employee attendance in real time.

Schedule management module. This module allows administrators and team leaders to manage employee work schedules.

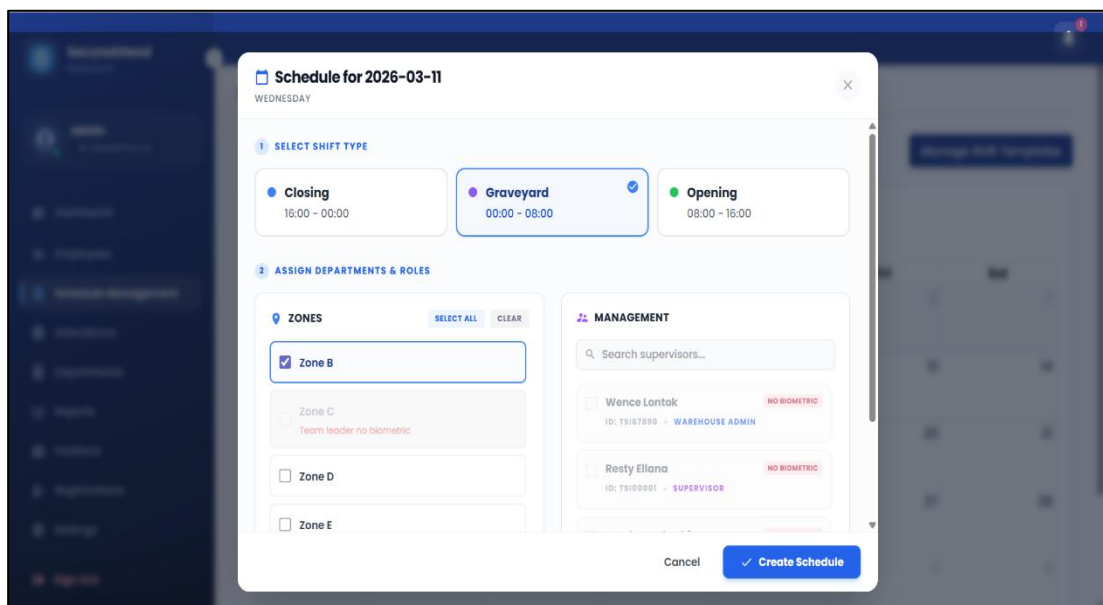


Figure 32. Screenshot of the schedule page of the system

This module allows administrators to create and manage employee work schedules within the system. Administrators can define shifts such as opening, closing, or graveyard schedules and assign them based on operational requirements. The module includes a calendar view that enables administrators to visualize employee

work assignments and identify potential scheduling conflicts. By organizing schedules through a centralized interface, administrators can efficiently manage workforce planning and ensure proper shift distribution.

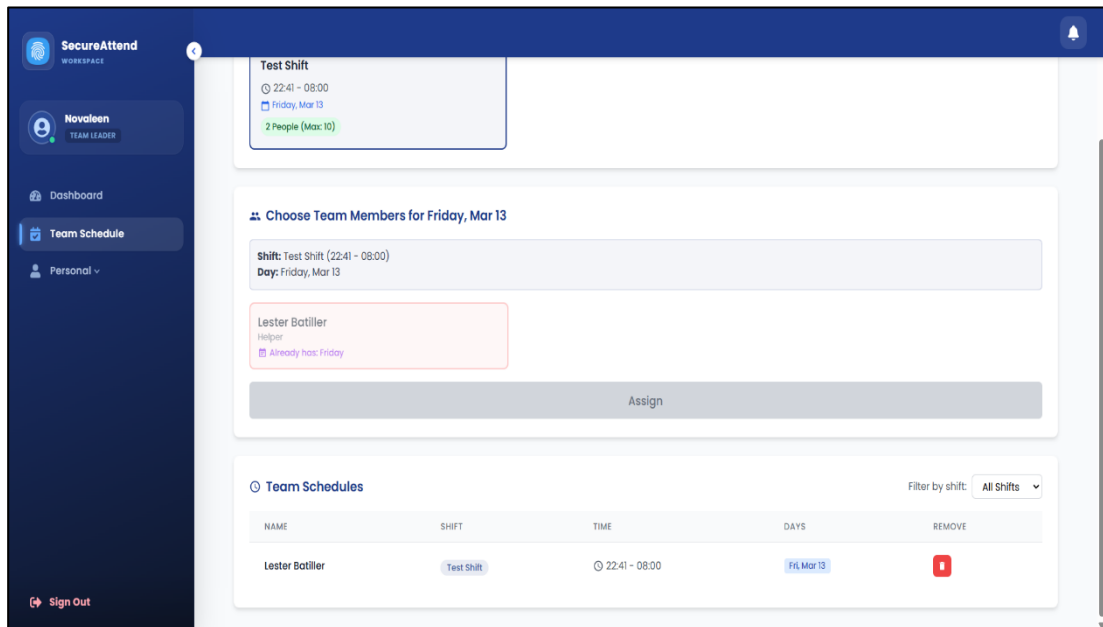


Figure 33. Screenshot of the team schedule page of the system

This module allows team leaders to assign employees to specific shifts within their designated teams. Through this interface, team leaders can select team members and assign them to scheduled shifts for particular dates based on operational needs. The system displays the selected shift schedule, including the shift time and assigned day, allowing team leaders to manage team assignments more effectively. It also enables team leaders to review existing team schedules and remove or update assignments when necessary. By providing an organized view of team schedules and assignment options, the module helps team leaders coordinate employee availability, ensure proper workforce coverage, and support smooth daily operations within their teams.

Department Module. This module allows administrators to organize employees according to their assigned departments or zones.

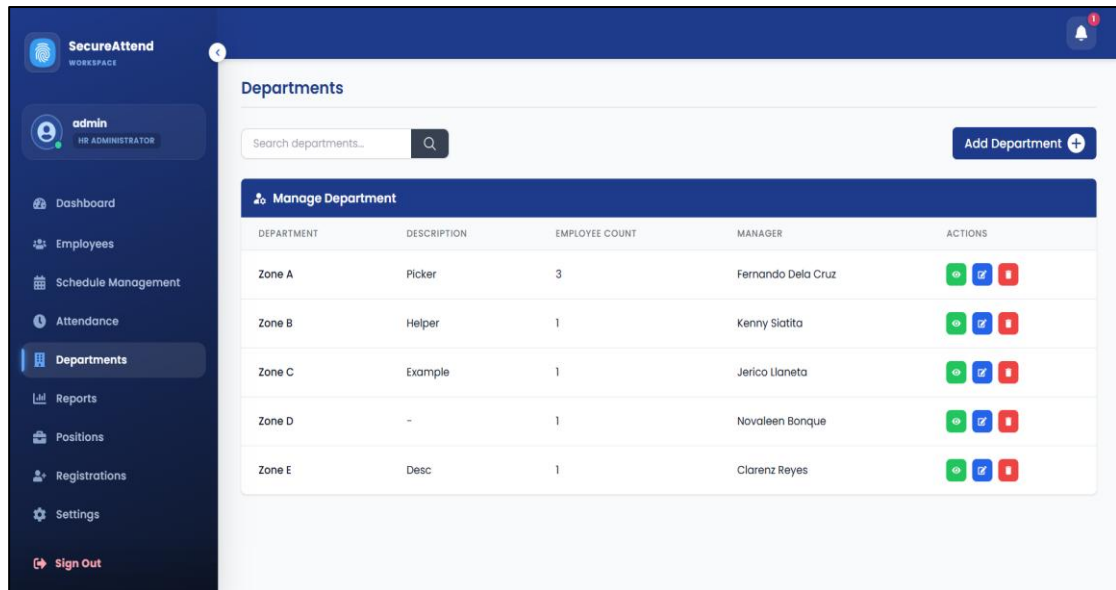


Figure 34. Screenshot of the department page of the system

This module enables administrators to manage the organizational departments or zones within the system. Administrators can create new departments, assign team leaders, and manage the employees assigned to each department. The module provides a structured list of departments including information such as department name, assigned supervisor or team leader, and the number of employees under each department. Administrators also have the ability to update department information or delete departments when necessary. This module helps maintain an organized structure within the organization and allows supervisors to effectively monitor attendance and schedules within their respective departments.

Notification and alerts Module. This module allows the system to send notifications and alerts to users regarding important system activities.

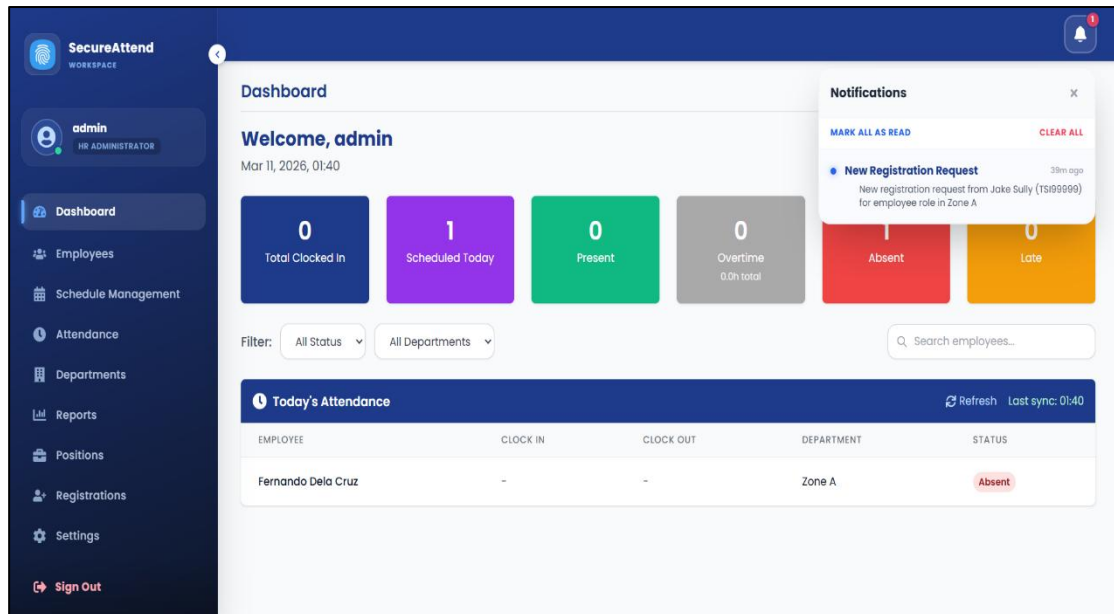


Figure 35. Screenshot of the notification page of the system

This module allows the system to automatically send notifications and alerts to users regarding important system activities. Notifications may include schedule updates, attendance confirmations, employee account requests, or other system-related announcements. These notifications are displayed within the system interface, allowing users to easily view important updates and system messages. By providing real-time alerts and notifications, the module improves communication between administrators, team leaders, and employees while ensuring that users remain informed about relevant system events and updates.

Reports Module. This module allows administrators to generate attendance reports for monitoring and payroll purposes.

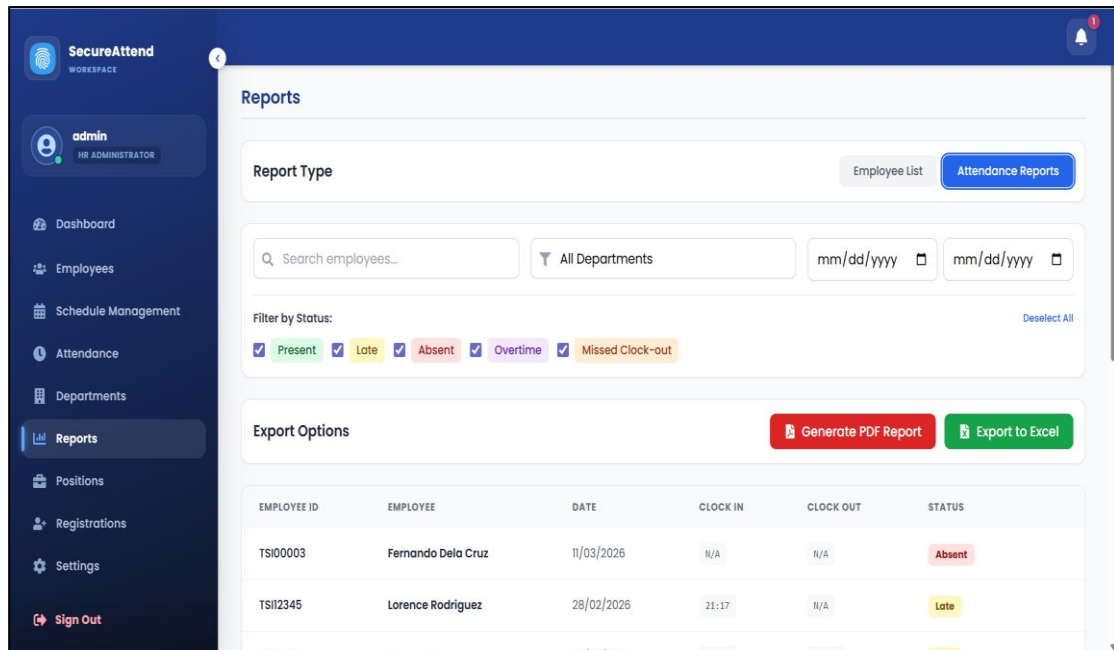


Figure 36. Screenshot of the reports page of the system

This module allows administrators to generate attendance reports based on the attendance data stored in the system database. The reports module provides filtering options that allow administrators to generate reports based on employee name, department, attendance status, or selected date range. The generated reports can be exported in formats such as Portable Document Format (PDF) or Microsoft Excel for documentation and payroll processing purposes. By automating the generation of attendance reports, the system significantly reduces the time and effort required in preparing attendance records while improving data accuracy and accessibility for administrative use.

System configuration module. This module allows administrators to configure important system settings.

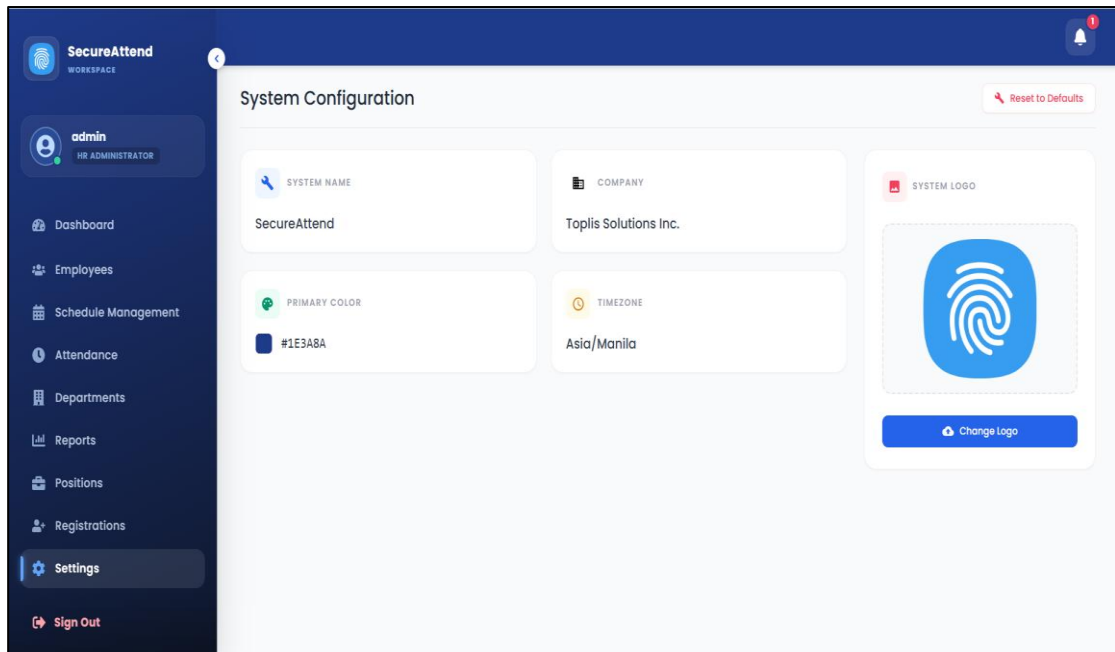


Figure 37. Screenshot of the system configuration page of the system

This module allows administrators to configure and customize important system settings according to the organization's requirements. Administrators can modify system information such as the system name, company name, system logo, and visual theme to align the system with the organization's branding. The module also allows administrators to manage certain system preferences that affect how the platform operates. Through these configuration options, the system can be customized to meet the operational needs of Toplis Solutions Inc. while maintaining consistency and proper system functionality.

System Development

SecureAttend: An Attendance Management System Using Biometrics for Toplis Solutions Inc. was developed using the Modified Waterfall Model, a method that offers a clear and organized process while still allowing flexibility when needed. This approach was selected because it helps the proponents complete each major phase of development systematically, while also making room for revisions whenever problems arise. The development process covered requirement analysis, system design, implementation, system testing, and operation and maintenance. Each phase was carefully documented and carried out in line with the objectives of the study.

Requirement Analysis. In this phase, the researchers gathered relevant information regarding the current attendance monitoring process of Toplis Solutions Inc. Data were collected through interviews with employees and supervisors, as well as through observation of the existing attendance procedures within the organization. The current attendance system relied on manual attendance recording using Daily Log Sheets (DLS), which resulted in time-consuming validation, human errors, and inefficient monitoring of employee attendance. Based on the collected data, the researchers identified the necessary system requirements needed for the development of SecureAttend. These requirements included biometric attendance recording, employee management, schedule management, department organization, notification alerts, and attendance report generation. The information gathered during this phase served as the foundation for designing the proposed system.

System Design. During this phase, the researchers designed the structure and workflow of the SecureAttend system. Several system diagrams were developed to illustrate the system architecture and processes, including use case diagrams, flowcharts, entity-relationship diagrams (ERD), and system wireframes. These design tools allowed the researchers to visualize how the different modules of the system would interact with one another. The system design focused on creating an organized

and user-friendly interface that allows administrators and employees to easily access system features. The modules designed for the system included user and employee management, attendance management, schedule management, department management, notification and alert functions, and report generation.

Implementation. In this phase, the researchers translated the system design into actual program code. The SecureAttend system was developed using modern web development technologies including React.js for the frontend interface, Node.js and Express.js for backend services, MySQL for database management, and JavaScript and C# for biometric integration. Each system module was coded and integrated into a centralized web-based platform that allows administrators and employees to access the system through a web browser. Special attention was given to the integration of the biometric fingerprint device to ensure accurate attendance recording. The implementation process focused on ensuring system functionality, data security, and smooth interaction between system modules.

System Testing. System testing was conducted to ensure that the SecureAttend system functions correctly and meets the system requirements identified during the earlier phases. The testing process involved verifying the functionality of each module, ensuring that biometric attendance recording works properly, and confirming that employee data, schedules, and reports are accurately processed within the system. The researchers also performed integration testing to ensure that all modules work together without errors. Any issues identified during testing were documented and corrected before proceeding to the deployment stage. This phase ensured that the system operates reliably and efficiently.

Operational and Maintenance. After the system successfully passed the testing phase, the SecureAttend system was deployed for actual use within Toplis Solutions Inc. The system was installed on a server environment that allows administrators and employees to access the platform through a web browser. The biometric fingerprint device was connected to the system to enable real-time

attendance recording. During deployment, the researchers conducted system demonstrations and training sessions to guide users on how to operate the system effectively. Maintenance and monitoring were also performed to ensure continuous system operation and to address any potential issues that may arise after implementation.

System Testing

This phase was conducted to verify that SecureAttend: An Attendance Management System Using Biometrics for Toplis Solutions Inc., Silang, Cavite operates as intended and meets its defined requirements. The testing focused on the test environment, and functionality, including biometric device integration for offline attendance logging, proper storage and synchronization of attendance records, and the smooth operation of the web-based platform across different operating systems, screen resolutions, browsers, and mobile devices. The results of the testing were used to identify errors, apply necessary improvements, and ensure that all system modules work properly before deployment in the actual workplace environment.

Table 1. Software test environment results.

INDICATOR	COMPLIANCE	REMARKS
1. The role of the tester had been clearly defined.	Complied	None
2. The tester used the project in Windows version:		
a. 10	Complied	None
b. 11	Complied	None
c. Android/IOS	Complied	None
d. Android/IOS	Complied	None
3. The tester used the project in the following screen resolutions:		

Table 2. Continued.

INDICATOR	COMPLIANCE	REMARKS
e. 720 x 1280	Complied	None
f. 1600 x 900	Complied	None
g. 1920 x 1080	Complied	None

The table shows that the system complied with all the required indicators. The tester's role was clearly defined, ensuring an organized evaluation process. The system was successfully tested on Windows 10, Windows 11, and Android/iOS, confirming compatibility across different platforms. It also functioned properly on various screen resolutions, including 720 × 1280, 1600 × 900, and 1920 × 1080. Overall, the results indicate that the system meets the required test environment standards and operates effectively across different devices and display settings.

Table 3. Software scenario environment results — Functionality of user and employee management module

INDICATOR	COMPLIANCE	REMARKS
1. Allow administrator to create, update, and deactivate user accounts.	Complied	None
2. Allow administrator to enroll fingerprint record for employees.	Complied	None
3. Allow administrator to assign departments and positions to employees.	Complied	None
4. Allow supervisors and team leaders to view employee information in their assigned departments	Complied	None

The table shows that the user and manager management module is working as intended. The system allows the administrator to create, update, and deactivate user accounts, which supports proper user management and access control. It also enables the administrator to enroll fingerprint records for employees, ensuring accurate

biometric attendance registration. In addition, the system allows the administrator to assign departments and positions to employees, which helps organize employee records properly. Lastly, the system allows supervisors and team leaders to view employee information in their assigned departments, making it easier to monitor and manage employees based on their respective roles. Overall, the results indicate that the system successfully met the required functional features and operated according to its intended purpose.

Table 4. Software scenario environment results — Functionality of the attendance management module

INDICATOR	COMPLIANCE	REMARKS
1. Allow employees to record attendance using biometric clock in and clock out.	Complied	None
2. Allow administrator, supervisor, and warehouse admin to view companywide attendance records.	Complied	None
3. Allow team leader to monitor department level attendance records.	Complied	None

The results presented in the table indicate that the attendance management module functions as intended. The system enables employees to record their attendance through biometric clock-in and clock-out features, thereby ensuring accurate and secure attendance tracking. It also allows the administrator, supervisor, and warehouse administrator to access company-wide attendance records, which facilitates efficient monitoring of attendance across the organization. Furthermore, the system enables the team leader to oversee attendance records at the departmental level, thereby supporting effective supervision of employees within their assigned department. Overall, the findings suggest that the system successfully fulfilled the required functional features and operated in accordance with its intended purpose.

Table 5. Software scenario environment results — Functionality of the schedule management module

INDICATOR	COMPLIANCE	REMARKS
1. Allow administrator and supervisor to create and manage schedule templates.	Complied	None
2. Allow administrator to assign schedules to employees.	Complied	None
3. Allow team leaders to assign schedules to their members within their assigned departments.	Complied	None
4. Allow employees to view their assigned work schedules.	Complied	None

The results on the table show that schedule management module works as intended. The system allows the administrator and supervisor to create and manage schedule templates, which supports proper scheduling setup and maintenance. It also allows the administrator to assign schedules to employees, ensuring that work schedules are properly distributed. In addition, team leaders can assign schedules to their members within their assigned departments, which helps manage scheduling at the departmental level. Lastly, the system allows employees to view their assigned work schedules, making it easier for them to stay informed of their work assignments. Overall, the results indicate that the schedule management module successfully met the required functional features and operated according to its intended purpose.

Table 6. Software scenario environment results — Functionality of the department management module

INDICATOR	COMPLIANCE	REMARKS
1. Allow administrator to create, update, and delete departments.	Complied	None
2. Allow administrator to assign team leaders to departments.	Complied	None
3 Allow supervisors to view department information and assign employees.	Complied	None

The findings for the department management module showed that the system met all the required indicators. It enables the administrator to create, edit, and remove departments, making it easier to organize and maintain department records properly. The system also allows the administrator to assign team leaders to specific departments, ensuring that each one has designated personnel responsible for supervision and coordination. Additionally, supervisors can view department details and the employees assigned to them, which helps in effectively monitoring the department structure and employee distribution. Overall, the results demonstrate that the department management module fulfilled the expected functional requirements and performed according to its intended purpose.

Table 7. Software scenario environment results — Functionality of the Reports module

INDICATOR	COMPLIANCE	REMARKS
1. Allow administrator to generate attendance reports for all employees in Excel or PDF format.	Complied	None
2. Allows administrator and supervisor to generate list of employees.	Complied	None

The results of the functionality of the reports module indicate that the system met all the requirements of the listed indicators. This is because the system can generate reports for all employees in the company by the administrator, thus ensuring efficient documentation and record-keeping. The system can generate reports in Excel or PDF format, thus ensuring efficiency in documentation and record-keeping. It can generate a list of employees for the administrator and the supervisor, thus ensuring efficiency in monitoring the employees. This indicates that the report's module met the required functional features of the system.

Table 8. Software scenario environment results — Functionality of the System configuration module

INDICATOR	COMPLIANCE	REMARKS
1 Allow administrator to modify system logo.	Complied	None
2. Allow administrator to modify system timezone.	Complied	None
3. Allow administrator to modify system name.	Complied	None
4. Allow administrator to modify company name.	Complied	None

The table shows that the System configuration module met all the defined requirements. The system enables the administrator to change the system logo, allowing for customization and branding. It also allows the administrator to adjust the system time zone, making sure the settings align with the organization's operational needs. Additionally, the administrator can update the system name and company name, helping keep the system identity and organizational details accurate. Overall, the findings show that the system configuration module successfully delivered the required functional features and performed according to its intended purpose.

Table 9. Software scenario environment results — Functionality of the Dashboard and notification module

INDICATOR	COMPLIANCE	REMARKS
1 Provide role specific dashboards for employees, supervisors, team leaders, and administrators.	Complied	None
2. Display attendance summaries and alerts on dashboards.	Complied	None
3. Send notifications for schedule updates and attendance related events.	Complied	None
4. Send notification for employee account request.	Complied	None

The results for the dashboard and notification module indicate that the system successfully met all the listed indicators. It provides role-based dashboards for employees, supervisors, team leaders, and administrators, allowing each user to access information relevant to their responsibilities. The dashboards display attendance summaries and alerts, helping users monitor important attendance details. In addition, the system sends notifications for schedule changes, attendance-related events, and employee account requests, which improves communication and coordination within the organization. Overall, the findings show that the module delivered the required functional features and performed according to its intended purpose.

Table 10. Software scenario environment results — Portability

DESKTOP AND MOBILE SPECIFICATION (MODEL, OS, RAM, RESOLUTION)	INSTALLED	WORKED PROPERLY	ACTIONS	RESPONSE TIME
Acer Nitro 5, Windows 11, 8GB, 1920x1080	N/A Browser	Yes	Test Attendance	1.2 s
HP ProDesk 400 G7, Windows 11, 8GB, 1920x1080	N/A Browser	Yes	Notification Test	1.3 s
Samsung Galaxy A12, Android 11, 16GB, 720x1600	N/A Browser	Yes	Create Reports	3.1 s
iPhone 13 Pro, iOS 15+, 6GB, 1170x2532	N/A Browser	Yes	Changing Logo	1.1 s
MacBook Air (M1, 2020), macOS Monterey, 8GB, 2560x1600	N/A Browser	Yes	Fingerprint Test	0.9 s
Samsung Galaxy S21, Android 12, 8GB, 1080x2400	N/A Browser	Yes	Attendance Monitor	2.0 s

The system was evaluated on several desktop and mobile devices to check its compatibility and overall performance. On an Acer Nitro 5 running Windows 11 with 8 GB of RAM and a 1920x1080 display, the attendance test was completed successfully in 1.2 seconds. An HP ProDesk 400 G7 with the same Windows 11 setup, 8 GB of RAM, and 1920x1080 resolution also handled the notification test properly, recording a response time of 1.3 seconds. On a Samsung Galaxy A12 running Android 11 with 16 GB of RAM and a 720x1600 resolution, the system successfully generated reports with a response time of 3.1 seconds. Meanwhile, an iPhone 13 Pro with iOS 15+, 6 GB of RAM, and a 1170x2532 resolution performed the logo-changing function correctly in 1.1 seconds. A MacBook Air (M1, 2020) running macOS Monterey with 8 GB of RAM and a 2560x1600 display showed the best performance during the

fingerprint test, achieving the fastest response time of 0.9 seconds. Finally, a Samsung Galaxy S21 with Android 12, 8 GB of RAM, and a 1080x2400 resolution was able to run the attendance monitor correctly with a response time of 2.0 seconds. Overall, these findings show that the system works well across a range of devices and operating systems, delivering dependable and efficient performance in every tested condition.

System Evaluation

The system evaluation of SecureAttend: An Attendance Management System Using Biometrics for Toplis Solutions Inc., Silang, Cavite was conducted to assess its overall performance, functionality, and usability. The product quality evaluation involved 80 respondents, consisting of 5 IT specialists, 3 domain experts, and 72 employees from the organization. On the other hand, the quality in use evaluation was assessed separately by 1 administrator. Responses were collected through survey questionnaires based on the ISO 25010 software quality model and were analyzed using a five-point Likert scale and weighted mean to determine the system's overall quality and effectiveness.

Product Quality. The product quality of SecureAttend was evaluated using the ISO 25010 product quality model. The evaluation focused on key software quality characteristics, namely functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. These criteria were used to determine whether the system can meet organizational requirements, perform efficiently, protect attendance data, and provide a reliable and user-friendly solution for managing employee attendance at Toplis Solutions Inc.

Table 11. Evaluation results for functional suitability criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	f	%
1. Functional completeness	0	0	0	0	12	15	30	37.5	38	47.5
2. Functional correctness	0	0	0	0	13	16.25	31	38.75	36	45
3. Functional appropriateness	0	0	0	0	15	18.75	34	42.50	31	38.75

Legend: f = frequency % = percentage

The findings for the functional suitability criterion show that the system was positively evaluated by the respondents. For functional completeness, most respondents rated the system at level 5 (47.5%), followed by level 4 (37.5%) and level 3 (15%), indicating that the system provides the necessary functions required by users. Similarly, functional correctness received strong ratings, with 45% of respondents giving a rating of 5, 38.75% giving a rating of 4, and 16.25% giving a rating of 3, suggesting that the system performs its functions accurately and as expected. For functional appropriateness, 38.75% of respondents rated the system at level 5, 42.50% rated it at level 4, and 18.75% rated it at level 3, indicating that the system's functions are relevant and suitable for user needs. Overall, the results indicate that the system achieved a high level of functional suitability, with most ratings concentrated on the higher levels and no negative responses recorded.

Table 12. Evaluation results for performance efficiency criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	f	%
4. Time behaviour	0	0	0	0	19	23.75	37	46.25	24	30
5. Resource utilization	0	0	0	0	17	21.25	38	47.5	25	31.25
6. Capacity	0	0	1	1.25	19	23.75	31	38.75	29	36.25

Legend: f = frequency % = percentage

The results for the performance efficiency criterion show that respondents gave the system a favorable evaluation. In terms of time behavior, most respondents rated

the system at level 4 (46.25%), followed by level 5 (30%) and level 3 (23.75), which indicates that it can perform its functions within an acceptable time frame. For resource utilization, 47.5% of respondents gave a rating of 4, 31.25% rated it as 5 and the other 21.25 rated it as level 3, showing that the system makes efficient use of available resources during operation. As for capacity, the majority rated the system at level 4 (38.75%), followed closely by level 5 (36.25%) and level 3 (23.75), with only 1.25% giving a rating of 2. This suggests that the system can generally handle the required workload and accommodate the expected number of users. Overall, the findings indicate that the system demonstrated good performance efficiency, as most responses were concentrated on the higher rating levels.

Table 13. Evaluation results for compatibility criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	f	%
7. Co-existence	0	0	0	0	16	20	36	45	28	35
8. Interoperability	0	0	0	0	19	23.75	28	35	33	41.25

Legend: f = frequency % = percentage

The evaluation results for the compatibility criterion shows a positive evaluation of the system by the respondents. For co-existence 16 respondents (20%) rated it at level 3, 36 respondents (45%) rated it at level 4, and 28 respondents (35%) rated it at level 5. This shows that the system can operate alongside other software and resources without conflict. For interoperability, 19 respondents (23.75%) rated it at level 3, 28 respondents (35%) rated it at level 4, and 33 respondents (41.25%) rated it at level 5. This indicates that the system can interact and exchange information effectively with other components or related technologies. Overall, the results show that the system achieved a good level of compatibility, since most of the responses were concentrated at levels 4 and 5, with no negative ratings recorded.

Table 14. Evaluation results for usability criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	f	%
9. Appropriateness	0	0	0	0	19	23.75	28	42.5	27	33.75
recognizability										
10. Learnability	0	0	0	0	20	25	29	36.25	31	38.75
11. Operability	0	0	0	0	13	16.25	39	48.75	28	35
12. User error protection	0	0	0	0	19	23.75	32	40	29	36.25
13. User interface	0	0	0	0	15	18.75	30	37.5	35	43.75
aesthetics										
14. Accessibility	0	0	0	0	14	17.5	34	42.5	32	40

Legend: f = frequency % = percentage

The evaluation results for the usability criterion demonstrate a favorable assessment of the system by the respondents. In terms of appropriateness recognizability, 19 respondents (23.75%) rated the system at level 3, 34 respondents (42.5%) at level 4, and 27 respondents (33.75%) at level 5. These findings indicate that the users were able to recognize the system as appropriate for their needs. Regarding learnability, 20 respondents (25%) rated the system at level 3, 29 respondents (36.25%) at level 4, and 31 respondents (38.75%) at level 5, suggesting that the system is easy to learn and understand. With respect to operability, 13 respondents (16.25%) rated it at level 3, 39 respondents (48.75%) at level 4, and 28 respondents (35%) at level 5, indicating that the system is easy to operate and control. For user error protection, 19 respondents (23.75%) rated it at level 3, 32 respondents (40%) at level 4, and 29 respondents (36.25%) at level 5, which suggests that the system effectively minimizes user errors during operation. In relation to user interface aesthetics, 15 respondents (18.75%) rated it at level 3, 30 respondents (37.5%) at level 4, and 35 respondents (43.75%) at level 5, indicating that the respondents found the interface visually appealing. Finally, for accessibility, 14 respondents (17.5%) rated it

at level 3, 34 respondents (42.5%) at level 4, and 32 respondents (40%) at level 5, demonstrating that the system is accessible and usable for its intended users. Overall, the results indicate that the system attained a high level of usability, as most responses were concentrated at levels 4 and 5.

Table 15. Evaluation results for reliability criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	f	%
15. Maturity	0	0	0	0	19	23.75	34	42.5	27	33.75
16. Availability	0	0	0	0	23	28.75	29	36.25	28	35
17. Fault tolerance	0	0	0	0	20	25	37	46.25	23	28.75
18. Recoverability	0	0	1	1.25	17	21.25	34	42.5	28	35

Legend: f = frequency % = percentage

The evaluation results for the reliability criterion indicate a positive assessment of the system by the respondents. For maturity, the highest rating was level 4 with 34 respondents (42.5%), followed by level 5 with 27 respondents (33.75%), and level 3 with 19 respondents (23.75%), showing that the system performs consistently during operation. For availability, the highest rating was level 4 with 29 respondents (36.25%), followed by level 5 with 28 respondents (35%), and level 3 with 23 respondents (28.75%), indicating that the system remains accessible and usable when needed. In terms of fault tolerance, the highest rating was level 4 with 37 respondents (46.25%), followed by level 5 with 23 respondents (28.75%), and level 3 with 20 respondents (25%), suggesting that the system can continue operating properly even when minor issues occur. For recoverability, the highest rating was level 4 with 34 respondents (42.5%), followed by level 5 with 28 respondents (35%), level 3 with 17 respondents (21.25%), and level 2 with 1 respondent (1.25%), which indicates that the system is generally capable of restoring data and resuming operation after failures. Overall, the results show that the system achieved a good level of reliability, since most of the responses were concentrated at levels 4 and 5.

Table 16. Evaluation results for security criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	f	%
19. Confidentiality	0	0	1	1.25	12	15	39	48.75	28	35
20. Integrity	0	0	0	0	14	17.5	37	46.25	29	36.25
21. Non-repudiation	0	0	0	0	12	15	40	50	28	35
22. Accountability	0	0	1	1.25	14	17.5	35	43.75	30	37.5
23. Authenticity	0	0	0	0	16	20	36	45	28	35

Legend: f = frequency % = percentage

The evaluation results for the security criterion indicate a positive assessment of the system by the respondents. For confidentiality, the highest rating was level 4 with 39 respondents (48.75%), followed by level 5 with 28 respondents (35%), level 3 with 12 respondents (15%), and level 2 with 1 respondent (1.25%), showing that the system is generally effective in protecting sensitive information from unauthorized access. For integrity, the highest rating was level 4 with 37 respondents (46.25%), followed by level 5 with 29 respondents (36.25%) and level 3 with 14 respondents (17.5%), indicating that the system can maintain the accuracy and consistency of data. In terms of non-repudiation, the highest rating was level 4 with 40 respondents (50%), followed by level 5 with 28 respondents (35%) and level 3 with 12 respondents (15%), suggesting that actions performed within the system can be traced and cannot easily be denied by users. For accountability, the highest rating was level 4 with 35 respondents (43.75%), followed by level 5 with 30 respondents (37.5%), level 3 with 14 respondents (17.5%), and level 2 with 1 respondent (1.25%), which indicates that the system can properly monitor and record user actions. Lastly, for authenticity, the highest rating was level 4 with 36 respondents (45%), followed by level 5 with 28 respondents (35%) and level 3 with 16 respondents (20%), showing that the system can verify the identity of users effectively. Overall, the results show that the system

achieved a good level of security, since most of the responses were concentrated at levels 4 and 5.

Table 17. Evaluation results for maintainability criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	f	%
24. Modularity	0	0	0	0	20	25	36	45	24	30
25. Reusability	0	0	1	1.25	17	21.25	37	46.25	25	31.25
26. Analysability	0	0	1	1.25	22	27.5	30	37.5	27	33.75
27. Modifiability	0	0	1	1.25	19	23.75	39	48.75	21	26.25
28. Testability	0	0	2	2.5	18	22.5	38	47.5	22	27.5

Legend: f = frequency % = percentage

The evaluation results for the maintainability criterion indicate a positive assessment of the system by the respondents. For modularity, the highest rating was level 4 with 36 respondents (45%), followed by level 5 with 24 respondents (30%) and level 3 with 20 respondents (25%), showing that the system is organized into components that can function effectively with minimal impact on one another. For reusability, the highest rating was level 4 with 37 respondents (46.25%), followed by level 5 with 25 respondents (31.25%), level 3 with 17 respondents (21.25%), and level 2 with 1 respondent (1.25%), indicating that parts of the system can still be reused for other functions or future improvements. In terms of analysability, the highest rating was level 4 with 30 respondents (37.5%), followed by level 5 with 27 respondents (33.75%), level 3 with 22 respondents (27.5%), and level 2 with 1 respondent (1.25%), suggesting that the system can be examined and understood when identifying issues or making improvements. For modifiability, the highest rating was level 4 with 39 respondents (48.75%), followed by level 5 with 21 respondents (26.25%), level 3 with 19 respondents (23.75%), and level 2 with 1 respondent (1.25%), which shows that the system can be modified without causing major problems to other components. Lastly, for testability, the highest rating was level 4 with 38 respondents (47.5%), followed by

level 5 with 22 respondents (27.5%), level 3 with 18 respondents (22.5%), and level 2 with 2 respondents (2.5%), indicating that the system can be effectively tested to confirm that it functions as expected. Overall, the results show that the system achieved a good level of maintainability, since most of the responses were concentrated at levels 4 and 5.

Table 18. Evaluation results for portability criterion

SUB-CRITERION	1		2		3		4		5	
	f	%	f	%	f	%	f	%	F	%
29. Adaptability	0	0	0	0	17	21.25	41	51.25	22	27.5
30. Installability	0	0	0	0	13	16.25	44	55	23	28.75
31. Replaceability	0	0	0	0	19	23.75	38	47.5	23	28.75

Legend: f = frequency % = percentage

The evaluation results for the portability criterion indicate a positive assessment of the system by the respondents. For adaptability, the highest rating was level 4 with 41 respondents (51.25%), followed by level 5 with 22 respondents (27.5%) and level 3 with 17 respondents (21.25%), showing that the system can be adjusted and used effectively in different operating environments. For installability, the highest rating was level 4 with 44 respondents (55%), followed by level 5 with 23 respondents (28.75%) and level 3 with 13 respondents (16.25%), indicating that the system can be installed and set up with ease. In terms of replaceability, the highest rating was level 4 with 38 respondents (47.5%), followed by level 5 with 23 respondents (28.75%) and level 3 with 19 respondents (23.75%), suggesting that the system can be replaced or substituted with minimal difficulty when necessary. Overall, the results show that the system achieved a good level of portability, since most of the responses were concentrated at levels 4 and 5.

Quality in Use. The Quality in Use evaluation of Secureattend will be answered by the warehouse administrator of Toplis Solutions, Inc. and focuses on key criteria including effectiveness, efficiency, and satisfaction, with emphasis on usefulness,

trust, pleasure, and comfort, as well as freedom from risk through economic, health and safety, and environmental risk mitigation, and context coverage in terms of completeness and flexibility.

Table 19. Evaluation results for quality in use

CRITERION	SCORE	INTERPRETATION
Effectiveness	5	Excellent
Efficiency	5	Excellent
Satisfaction		
Usefulness	5	Excellent
Trust	5	Excellent
Pleasure	5	Excellent
Comfort	5	Excellent
Freedom from Risk		
Economic Risk Mitigation	5	Excellent
Health and Safety Risk Mitigation	5	Excellent
Environmental Risk Mitigation	5	Excellent
Context Coverage		
Context Completeness	5	Excellent
Flexibility	4	Very Good

The evaluation results for quality in use show that the system performed excellently across all measured aspects. Respondents rated effectiveness and efficiency at the highest score, indicating that the system helps users achieve their goals quickly and accurately. All sub-criteria under satisfaction, including usefulness,

trust, pleasure, and comfort, were also rated excellent, showing that users have a positive and satisfying experience when using the system. Similarly, freedom from risk in economic, health and safety, and environmental aspects received the highest ratings, demonstrating that the system minimizes potential risks during its operation. Lastly, context coverage, including completeness was rated excellent and the flexibility was rated very good, indicating that the system works effectively in various situations and user contexts. Overall, the system achieved a high level of quality in use.

Implementation Activities and Plans

The implementation of SecureAttend will be carried out in several steps to ensure that the system is properly introduced and utilized within the organization. The process begins with obtaining approval from the organization administrator, followed by installing and configuring the system, distributing user manuals, conducting orientation and training for users, and monitoring the system after deployment. This implementation plan aims to guide users in operating the system effectively, prevent possible issues during usage, and ensure the continuous and efficient operation of the SecureAttend system. The table below presents the implementation activities, persons involved, and the duration required for each activity.

Table 20. Implementation plan

STRATEGY	ACTIVITIES	PERSONS INVOLVED	DURATION
Approval from the organization administrator	Preparation and submission of necessary documents for system implementation approval	Researchers, organization administrator	1 day
System installation and setup	Installation of the system, database configuration, biometric device setup	Researchers, organization administrator, IT personnel	2 days
System configuration	Setting up system settings including company information, employee accounts, departments, and schedules	Researchers, organization administrator	1 day
Information distribution	Preparation and distribution of user manuals and documentation for the system usage	Researchers, organization administrator, employees	1 day
One-day user orientation	Introduction and demonstration of the system features and function	Researchers, organization administrator, employees	1 day
System training	Hands-on training for administrators and employees on attendance recording, schedule viewing, and report generation	Researchers, Organization administrator, employees	3 day
System testing and evaluation	Monitoring system performance and collecting feedback using ISO 25010 evaluation instruments	Researchers, Organization administrator	Twice a month for 3 months
Maintenance and technical support	Monitoring system performance, fixing possible errors, and providing technical assistance	Researchers	6 months

The table in the implementation plan presents a step-by-step process for deploying the SecureAttend: An Attendance Management System Using Biometrics for Toplis Solutions Inc., Silang, Cavite, ensuring that the system is properly introduced, utilized, and maintained within the organization. The process begins with obtaining formal approval from the organization administrator through the preparation and submission of the necessary documents required for system implementation. This step confirms the organization's authorization to proceed with the deployment of the system. After approval is granted, the system installation and setup are conducted, which include installing the system, configuring the database, and setting up the biometric fingerprint device required for attendance recording. This ensures that the system environment is properly prepared for operational use. The next phase involves configuring the system settings, including company information, employee accounts, departments, and schedules, to ensure that the system reflects the organizational structure of Toplis Solutions Inc. Following this step, user manuals and documentation are prepared and distributed to guide administrators and employees in understanding the system features and functions. A one-day orientation is then conducted to introduce the system and demonstrate its basic operations. This is followed by a three-day training program that provides hands-on instruction for administrators and employees on how to properly use the system for attendance recording, schedule viewing, and report generation. After the training, the system undergoes monitoring and evaluation to assess its performance and gather feedback from users using the ISO 25010 evaluation instruments over a three-month period. Finally, maintenance and technical support are provided for six months to monitor system performance, resolve potential issues, and assist users when necessary. Overall, this implementation approach ensures that the SecureAttend system is properly deployed, evaluated, and maintained to effectively support the attendance management needs of the organization.

SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter presents the summary of the study, the conclusions drawn from the results of the system evaluation, and the recommendations for future improvements of the SecureAttend system.

Summary

This study developed SecureAttend: An Attendance Management System Using Biometrics for Toplis Solutions Inc., Silang, Cavite to improve the organization's attendance monitoring process. The system was designed to address the problems associated with manual attendance recording such as human errors, time-consuming validation, and the possibility of buddy punching. The study aimed to develop a web-based attendance management system integrated with biometric fingerprint authentication to ensure accurate, reliable, and secure attendance recording.

The development of the system followed the Modified Waterfall Model, which included the phases of requirement analysis, system design, implementation, system testing, and deployment and maintenance. During the requirement analysis phase, the researchers gathered information about the existing attendance process through interviews and observations. The system design phase involved the creation of system diagrams and module structures to illustrate how the system would function. The implementation phase translated the design into actual program code using technologies such as React.js, Node.js, Express.js, MySQL, JavaScript, and C# for biometric integration. System testing was conducted to ensure that all modules of the system function properly across different devices, operating systems, and screen resolutions.

The developed system includes several modules such as user and employee management, attendance management, schedule management, department management, reports, notifications and alerts, and system configuration. These

modules work together to provide an automated and efficient attendance monitoring solution for the organization.

The system was evaluated using the ISO/IEC 25010 software quality model with 80 respondents composed of IT specialists, domain experts, the organization administrator, and employees. The evaluation results showed that the system achieved favorable ratings across product quality characteristics including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Additionally, the quality in use evaluation obtained excellent results in effectiveness, efficiency, satisfaction, freedom from risk, and context coverage, indicating that the system provides a reliable and user-friendly solution for attendance management.

Conclusion

Based on the results of the system development, testing, and evaluation, the following conclusions were drawn:

1. The SecureAttend system successfully automated the attendance monitoring process of Toplis Solutions Inc. by integrating biometric fingerprint authentication with a web-based platform, which improved the accuracy and reliability of attendance recording.
2. The system successfully implemented the required functional modules including employee management, attendance monitoring, scheduling, department management, reporting, notifications, and system configuration, allowing administrators and employees to manage attendance-related activities efficiently.
3. The results of the ISO/IEC 25010 evaluation indicate that the system achieved high ratings in product quality characteristics such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, demonstrating that the system meets the required quality standards

4. The quality in use evaluation showed that the system performed excellently in terms of effectiveness, efficiency, user satisfaction, freedom from risk, and context coverage, indicating that the system effectively supports users in performing attendance management tasks.
5. Overall, the SecureAttend system proved to be a reliable and effective attendance management solution that can improve attendance monitoring efficiency, reduce manual errors, and support better workforce management within Toplis Solutions Inc.

Recommendation

Based on the findings of the study, the following recommendations are proposed to further enhance the SecureAttend system:

1. Future researchers may improve the system by integrating additional biometric technologies such as facial recognition to provide alternative attendance authentication methods.
2. The system may be enhanced by adding real-time mobile notifications or alerts to inform employees and administrators about attendance updates, schedule changes, or important system announcements.
3. Future development may include the integration of payroll computation features based on attendance records for salary monitoring.
4. The system may also be enhanced by implementing advanced analytics and reporting features that provide detailed insights into employee attendance patterns and workforce performance.

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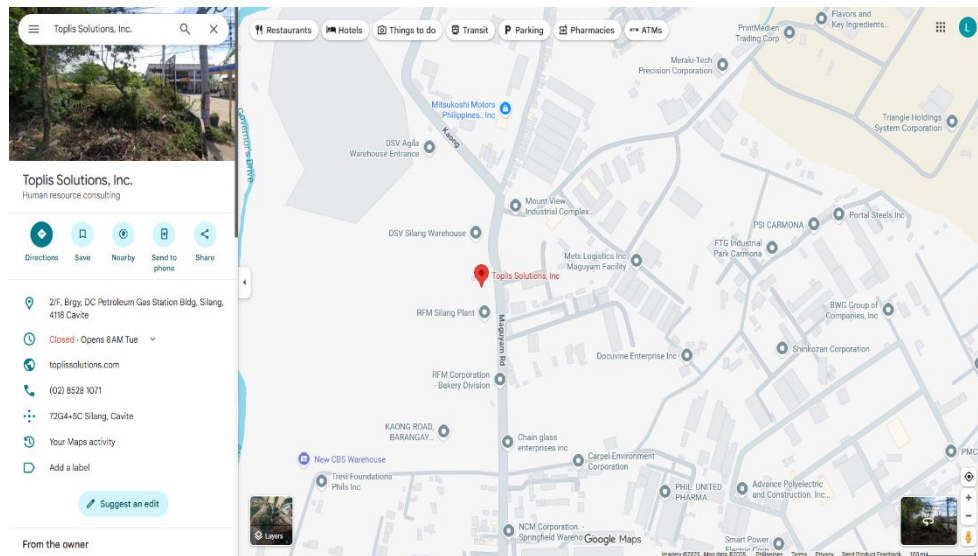
APPENDIX FIGURES



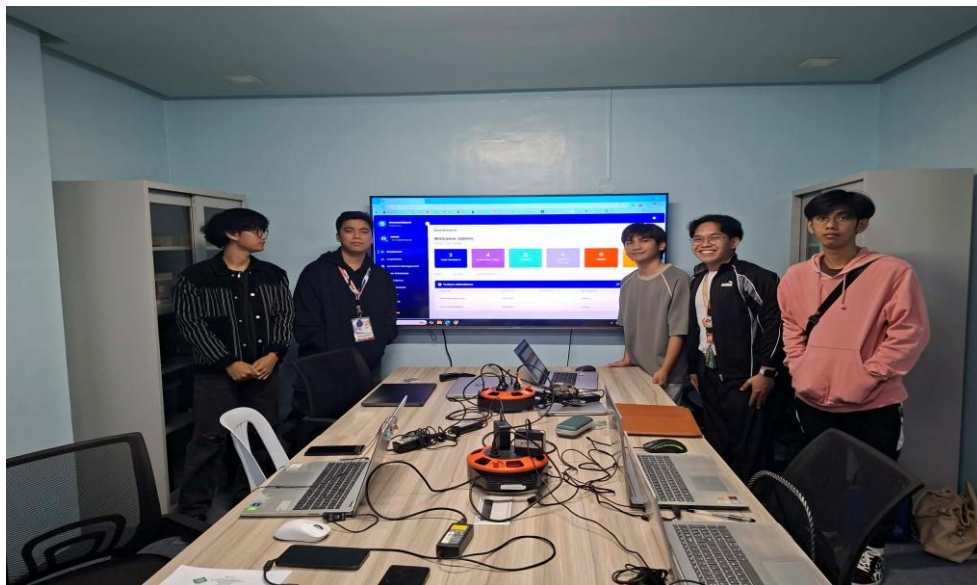
Appendix Figure 1. Toplis Solution, Inc.



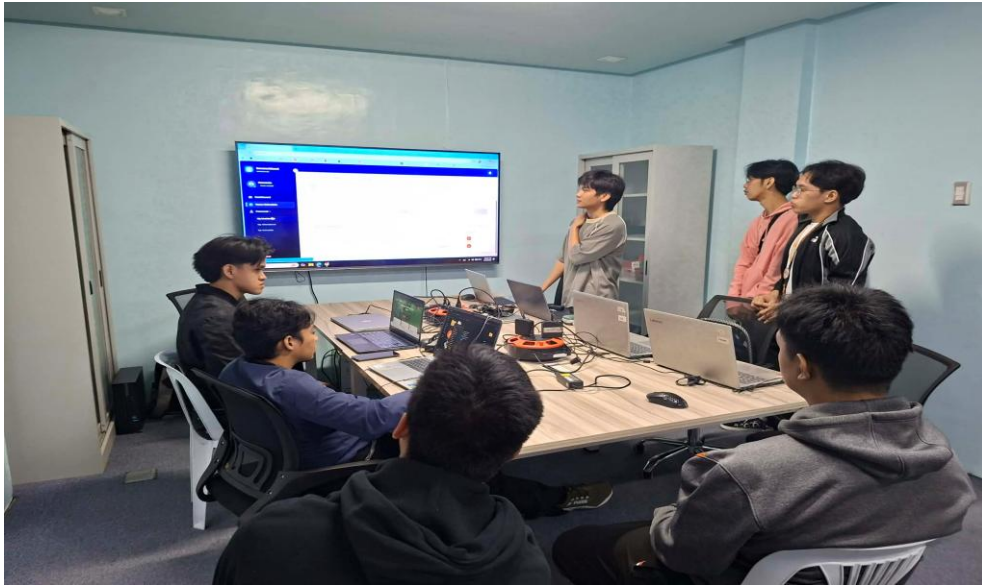
Appendix Figure 2. Alfamart Warehouse



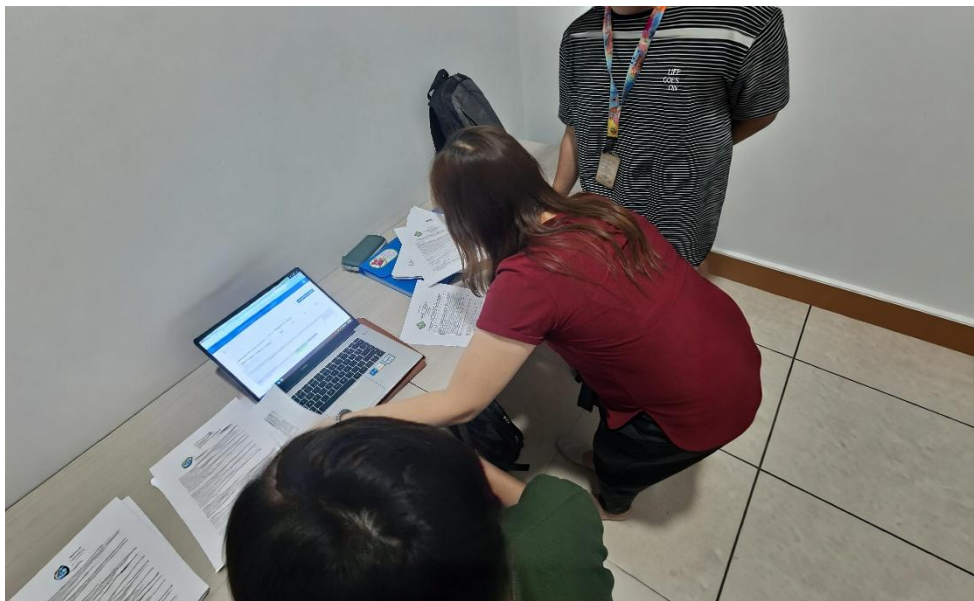
Appendix Figure 3. Toplis Soluton, Inc Map Location



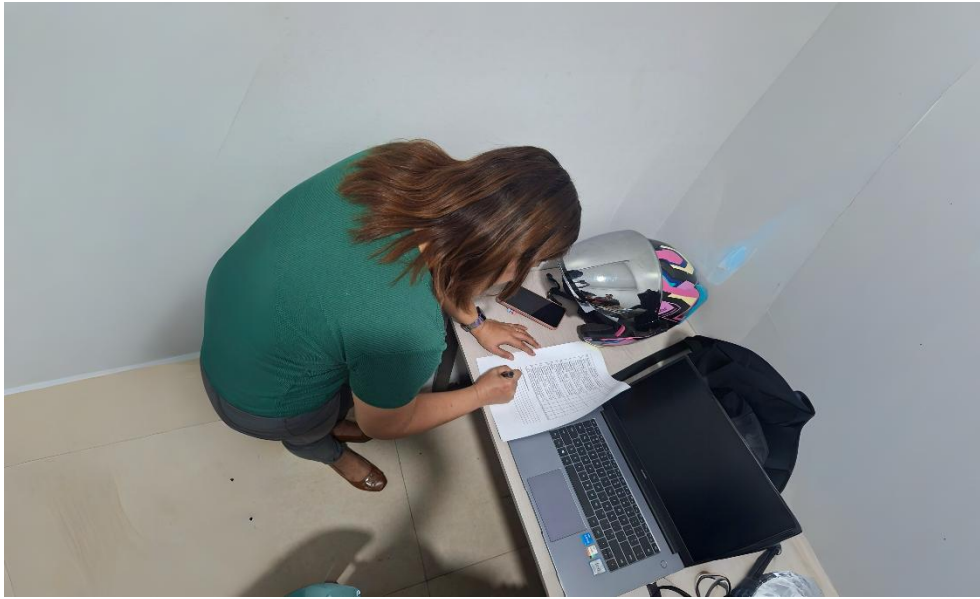
Appendix Figure 4. System Evaluation of Product Quality for IT Expert



Appendix Figure 5. System Evaluation of Product Quality for IT Expert



Appendix Figure 6. System Evaluation of Product Quality for Domain Expert



Appendix Figure 7. System Evaluation of Product Quality for Domain Expert



Appendix Figure 8. System Evaluation of Quality in Use

APPENDICES

Appendix 1



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DEPARTMENT OF INDUSTRIAL AND INFORMATION TECHNOLOGY

Technical Feasibility Assessment

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Program : **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

Title of Study: : **SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE**

Project Overview

SecureAttend: An Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite is an attendance system that helps in managing the attendance of employees of Toplis Solutions inc. Maguyam, Silang, Cavite. With added benefit of biometric technology for enhance security. The goal of the system is to enhance the accuracy of the attendance and improve the efficiency of the attendance process.

The project's functional requirements are user registration and management, attendance check in/check out, time tracking, admin dashboard, employee information, attendance sheet, notification and alerts. While the non-functionals requirements are security, usability, and system reliability.

The project will be completed in two semesters, from requirements gathering to design, development, testing, and deployment, with the expenses being distributed largely for web hosting and Email services, using open-source technology to ensure that costs remain at an absolute minimum.

Technology Assessment

Existing Technologies:

- a. The Toplis Solutions Inc. currently uses a manual attendance process through the use attendance slips provided by the organization. This manual process is prone to human error which affects the accuracy of the attendance. Applicable technologies for SecureAttend attendance system are:
 - Database Management Systems (DBMS) – MySQL to manage the structured data (employee’s information, and attendance record).
 - Web Development Frameworks – React.js, Express.js, and Node.js, to develop the frontend and backend components of the system.
 - Software Development – C# to develop the software that will be use in biometric integration
 - Biometric Technologies – fingerprint scanner for employee’s attendance Validation
- b. The chosen technologies are tried, tested, and commonly used in today's attendance management system. MySQL is a tried and tested relational database management system that offers data integrity, scalability, and structured storage, and is well-suited to manage the employee’s attendance data. React.js and Node.js offer a lean and scalable web development platform with high-performance delivery, adaptability, and responsiveness. is ideal for real-time verification and large-scale operations. Lastly Biometrics is great for attendance validation and verification. Overall, these technologies are suitable for the project’s requirements.

Emerging Technologies:

- a. Some emerging technologies can improve project’s functionality, security, and efficiency. These are:
 - Cloud-Hosting – Platforms like AWS, Firebase, or DigitalOcean provide scalable and reliable hosting for the system.
 - Connectivity – With faster network speeds, 5G can potentially support better real-time system performance and accessibility,
- b. Web hosting such as AWS and Firebase are widespread and well established and could be used to make the system more scalable and flexible. Additionally, the rise of 5G technology can be utilize, but due to the fact that it is not yet operational everywhere, its practical usefulness is questionable. All of these emerging technologies have great potential in improving the overall performance of the system.

Resource Assessment

Hardware:

- a. The hardware requirements of SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite are the following:
 - Servers or Web Hosting – a web hosting platform (e.g. Namecheap, and Hostinger) are used to host the project
 - Workstations (Desktops/Laptops) – Used by programmers for development, testing, and maintenance of all systems. Also required by the organization to manage the attendance, schedule and system configurations.

- Mobile Devices (Smartphones/Tablets) – are used by the employees access the website and check their schedule
 - Network Infrastructure (Strong Internet Connection) – Needed to host the system, synchronize information, and facilitate easy access to the web application.
 - Biometric Scanner – is used by the employees to validate their attendance
- b. Available workstations (laptops and desktops) for the admin are enough to access the system, though the system requires a secure server to save all of the project and organization data. A secured and high-speed network infrastructure which allows for real-time access to the system is provided by the organization themselves.

Software:

- a. The software requirements of SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite are the following:
- Operating System: Windows 10 or 11 (for development and administrative use).
 - Programming Languages: JavaScript (for both frontend and backend development), and C# for hardware integration.
 - Development Frameworks & Libraries: React.js for building the frontend user interface.
 - Node.js: For managing backend.
 - Database Management System: MySQL for organizing structured data storage and management.
 - Integrated Development Environment: Visual Studio Code – For coding, debugging, editing, and system development
- b. Windows 10 or 11 are the most used operating system as of today and most of the software that will be used in the development of the project no longer support older version. Hence the it is the most important software requirements for the project. React.js, Node.js, C# and MySQL are critical software required for this project. They provide the ability to develop the project easily and manage data efficiently. These technologies help handle large amounts of information. Visual Studio Code IDE is an excellent programming tool that is available for free. This will be used as the primary application in the development of the project.

Personnel:

- a. The technical skills required to develop the SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite are the following:
- Frontend Development: Knowledge of React.js, HTML, CSS, and JavaScript are necessary to develop a user-friendly website that works well on all devices.
 - Backend Development: It is vital for the backend of the application to build and manage through either Node.js server.
 - Hardware Integration: Knowledge in C# is necessary for biometric integration.

- Database Management: Knowledge in MySQL ensures that the appropriate procedures in storing and retrieving data safely exist.
 - Security and Authentication: Grasping concepts like role-based access control, data encryption. These skills will be beneficial in keeping the safety of user data and limiting its access to authorized people only. To maintain trust and privacy in any system, protection of personal information and user verification will be done with the help of these security measures.
- b. The development team are consisting of BSIT students of Cavite State University Carmona Campus. Most members are proficient in JavaScript, React.js, Node.js, MySQL, and also familiar with web-based platforms. This knowledge allows them to build both the frontend, and backend. Noted that some have knowledge in basic API integration. But further research and testing are needed to implement Biometric Integration. Possibilities of requiring external resources and mentors exist for expertise in system-security measures. Overall, the team has the needed skills for developing the project but still requires guidance from said mentors might be needed to advance some security and deployment features.

Risk Assessment

The development of a SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite is also prone to some technical risks that might affect the development of the project. The largest risks include system downtime and crashing of a server rendering the system unavailable, in case of downtime of the selected web hosting service (Namecheap and Hostinger). To counter this, server backup will be in place. Another problem that needs to be consider are hardware malfunction caused by many variables. For this a backup method will be in place

Security and data privacy threats are also high priority, as unauthorized access to attendance information and any sensitive information. To mitigate this risk, the system will adopt RBAC (Role Based Access Control), data encryption. Another possible risk is compatibility and browser issues. For a seamless user experience, cross-browser testing and responsive design will be implemented. Finally, gaps in technical knowledge may lead to problems in integrating the biometric technology. This risk will be addressed through extra training, research, expertise consultation and a hands-on risk mitigation.

Overall Technical Feasibility Assessment

Based on the assessment, SecureAttend: Attendance Management System using Biometrics for Toplis Solutions Inc., Silang, Cavite is technically viable in terms of possessing the necessary skills and technologies required for the project. Though there are still some questionable things that needs to be considered especially with the technical knowledge required in integrating the MFA technologies. But overall, the project is technically feasible with the right planning, resources, and expertise.

Appendix 2



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DEPARTMENT OF INDUSTRIAL AND INFORMATION TECHNOLOGY

Budgetary Requirements

Name of Researcher(s) : **MICHAEL JHEY P. ABUEDO, MARK VINCENT T. CABALLES, MEINARD C. LANGUITA, and LORENCE R. RODRIGUEZ**

Program : **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

Title of Study: : **SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE**

PARTICULARS	AMOUNT (P)
1. Biometric Module	2000.00
2. Hosting Fee	2500.00
3. Miscellaneous Fee	2000.00
TOTAL	6500.00

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Gantt Chart

Title of Study: : **SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE**

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Appendix 4



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DEPARTMENT OF INDUSTRIAL AND INFORMATION TECHNOLOGY

I. INTRODUCTION

The SecureAttend is an attendance system with biometric technology for Toplis Solutions Inc. Maguyan, Silang, Cavite. It is developed in hopes of replacing the manual attendance process while following the institution's policies.

II. OBJECTIVES

The main objective of the study is this test document is to discover the bugs and fix the issues before the user evaluation.

This project will help Toplis Solutions Inc Maguyam, Silang, Cavite branch.

Specifically, it aims to:

1. develop test cases for the testers (IT experts);
2. test the system in terms of functionality, performance and usability; and
3. gather feedback from the testers.

III. TASKS

The testers were provided a questionnaire which includes a checklist of attributes that the proposed system should comply with. A column for the tester's comments or recommendations is also provided.

The bugs discovered in this tool were addressed by the developer if deemed necessary and/or within the requirements and specifications of this project.

IV. HARDWARE REQUIREMENTS

The following hardware specifications ensure that the system runs. Before the start of the software testing, the minimum hardware requirements as stated below is compared to the actual specification used by the tester.

Table 1. Minimum hardware specification

HARDWARE SPECIFICATION	MINIMUM
Browser	Google Chrome, Apple Safari & Opera
Mobile Screen Resolution	375x667
Desktop Screen resolution	1024x768

V. SOFTWARE TESTING QUESTIONNAIRE

Table 2. Test environments

ITEM(S) FOR CHECKING	COMPLIANCE	REMARKS (IF ANY)
1. The role of the tester had been clearly defined.		
2. The tester used the project in Operating system version: a. Windows 10 and 11 b. MacOS c. Android/IOS		
3. The tester used the project in the following screen resolutions: a. 720x1280 b. 1600x900 c. 1920x1080		

Test Scenarios:

Table 3. Functionality

CRITERION	COMPLIANCE	REMARKS/ REQUEST FOR CHANGE (IF ANY)
A. User and Employee Management Module		
1. Allow administrator to create, update, and deactivate user accounts.		
2. Allow administrator to enroll fingerprint record for employees.		
3. Allow administrator to assign departments and positions to employees.		
4. Allow supervisors and team leaders to view employee information in their assigned departments.		
B. Attendance Management Module		
1. Allow employees to record attendance using biometric clock in and clock out.		
2. Allow administrator, supervisor, and warehouse admin to view companywide attendance records.		
3. Allow team leader to monitor department level attendance records.		

CRITERION	COMPLIANCE	REMARKS/ REQUEST FOR CHANGE (IF ANY)
C. Schedule Management Module		
1. Allow administrator and supervisor to create and manage schedule templates.		
2. Allow administrator to assign schedules to employees.		
3. Allow team leaders to assign schedules to their members within their assigned departments.		
4. Allow employees to view their assigned work schedules.		
D. Department Management Module		
1. Allow administrator to create, update, and delete departments.		
2. Allow administrator to assign team leaders to departments.		
3. Allow supervisors to view department information and assigned employees.		
E. Reports Module		
1. Allow administrator to generate attendance reports for all employees in Excel or PDF format.		
2. Allows administrator and supervisor to generate list of employees.		
F. System Configuration Module		
1. Allow administrator to modify system logo.		
2. Allow administrator to modify system timezone.		
3. Allow administrator to modify system name.		
4. Allow administrator to modify company name.		
G. Dashboard and Notification Module		
1. Provide role specific dashboards for employees, supervisors, team leaders, and administrators.		

CRITERION	COMPLIANCE	REMARKS/ REQUEST FOR CHANGE (IF ANY)
2. Display attendance summaries and alerts on dashboards.		
3. Send notifications for schedule updates and attendance related events.		
4. Send notification for employee account request.		

Table 4. Portability

DESKTOP & MOBILE SPECIFICATIONS	INSTALLED	WORKED PROPERLY	ACTIONS	RESPONSE TIME(S)
(MODEL, OS, RAM, RESOLUTION)				
Acer Nitro 5, Windows 11, 8GB, 1920x1080				
HP ProDesk 400 G7, Windows 11, 8GB, 1920x1080				
Samsung Galaxy A12, Android 11, 16GB, 720x1600				
iPhone 13 Pro, iOS 15+, 6GB, 1170x2532				
MacBook Air (M1, 2020), macOS Monterey, 8GB, 2560x1600				
Samsung Galaxy S21, Android 12, 8GB, 1080x2400				

Prepared by:

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MARK VINCENT T. CABALLES,
MEINARD C. LANGUITA,
LORENCE R. RODRIGUE.

Appendix 5



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EVALUATION INSTRUMENT FOR SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE

Dear Participant,

Good day! We conducting research entitled **SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE** “. In line with this, we respectfully request your assistance in filling out this evaluation form. It will not be a problem if you wish not to participate, but your responses will be highly valued. The evaluation form can be completed anonymously. Responses from completed questionnaires will be collated for analysis; once complete, the original questionnaires will be kept electronically. Rest assured that all information indicated therein will be treated with utmost confidentiality under the Data Privacy Law of 2012 and strictly used only for the above purpose. All the gathered information/data will also be retained in the system and used as a part of the historical data for further analysis. If you wish to learn more about the results of the research, please send an email to cc.lorence.rodriguez@cvsu.edu.ph.

We are hoping for your kind consideration and support. Thank you very much.

Name (Optional): _____ Date: _____
Address (Optional): _____
Profession: _____ Specialization: _____

☐ IT expert

☐ Domain Expert

☐ Secondary Users

Instructions: Please evaluate using the given scale and placing a checkmark (✓) on the appropriate column corresponding to your response.

Numerical Rating:

5 – Excellent

4 – Very Good

3 – Good

2 – Fair

1 – Poor

INDICATOR	5	4	3	2	1
Functional Suitability					
1. Functional completeness - Degree to which the set of functions covers all the specified tasks and user objectives.					

INDICATOR	5	4	3	2	1
2. Functional correctness - Degree to which a product or system provides the correct results with the needed degree of precision.					
3. Functional appropriateness - Degree to which the functions facilitate the accomplishment of specified tasks and objectives.					
Performance Efficiency					
4. Time behaviour - Degree to which the response and processing times and throughput rates of a product or system, when performing its functions, meet requirements.					
5. Resource utilization - Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.					
6. Capacity - Degree to which the maximum limits of a product or system parameter meet requirements.					
Compatibility					
7. Co-existence - Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
8. Interoperability - Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.					
Usability					
9. Appropriateness recognizability - Degree to which users can recognize whether a product or system is appropriate for their needs.					
10. Learnability - Degree to which a product or system can be used by specified users to achieve specified goals of learning to use the product or system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
11. Operability - Degree to which a product or system has attributes that make it easy to operate and control.					
12. User error protection - Degree to which a system protects users against making errors.					

INDICATOR	5	4	3	2	1
13. User interface aesthetics - Degree to which a user interface enables pleasing and satisfying interaction for the user.					
14. Accessibility - Degree to which a product or system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
Reliability					
15. Maturity - Degree to which a system, product or component meets needs for reliability under normal operation.					
16. Availability - Degree to which a system, product or component is operational and accessible when required for use.					
17. Fault tolerance - Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.					
18. Recoverability - Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.					
Security					
19. Confidentiality - Degree to which a product or system ensures that data are accessible only to those authorized to have access.					
20. Integrity - Degree to which a system, product or component prevents unauthorized access to, or modification of, computer programs or data.					
21. Non-repudiation - Degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.					
22. Accountability - Degree to which the actions of an entity can be traced uniquely to the entity.					
23. Authenticity - Degree to which the identity of a subject or resource can be proved to be the one claimed.					
Maintainability					
24. Modularity - Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.					

INDICATOR	5	4	3	2	1
25. Reusability - Degree to which an asset can be used in more than one system, or in building other assets.					
26. Analysability - Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.					
27. Modifiability - Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.					
28. Testability - Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.					
Portability					
29. Adaptability - Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.					
30. Installability - Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.					
31. Replaceability - Degree to which a product can replace another specified software product for the same purpose in the same environment.					

Adopted from the International Organization for Standardization (ISO) 25010 for product quality

Remarks/Comments/Suggestions:

Signature of Respondent

Appendix 6



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EVALUATION INSTRUMENT FOR SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE

Dear Participant,

Good day! We conducting research entitled **SECUREATTEND: AN ATTENDANCE MANAGEMENT SYSTEM USING BIOMETRICS FOR TOPLIS SOLUTIONS INC., SILANG, CAVITE** “. In line with this, we respectfully request your assistance in filling out this evaluation form. It will not be a problem if you wish not to participate, but your responses will be highly valued. The evaluation form can be completed anonymously. Responses from completed questionnaires will be collated for analysis; once complete, the original questionnaires will be kept electronically. Rest assured that all information indicated therein will be treated with utmost confidentiality under the Data Privacy Law of 2012 and strictly used only for the above purpose. All the gathered information/data will also be retained in the system and used as a part of the historical data for further analysis. If you wish to learn more about the results of the research, please send an email to cc.lorence.rodriguez@cvsu.edu.ph.

We are hoping for your kind consideration and support. Thank you very much.

Name (Optional): _____ Date: _____
 Address (Optional): _____
 Profession: _____ Specialization: _____

Instructions: Please evaluate using the given scale and placing a checkmark (✓) on the appropriate column corresponding to your response.

Numerical Rating:

5 – Excellent 4 – Very Good 3 – Good 2 – Fair 1 – Poor

INDICATOR	5	4	3	2	1
Effectiveness					
1. Effectiveness - Degree of accuracy and completeness with which users achieve specified goals.					
Efficiency					

INDICATOR	5	4	3	2	1
2. Efficiency - Degree of resources expended in relation to the accuracy and completeness with which users achieve goals.					
Satisfaction					
3. Usefulness - Degree to which a user is satisfied with their perceived achievement of pragmatic goals, including the results of use and the consequences of use.					
4. Trust - Degree to which a user or other stakeholder has confidence that a product or system will behave as intended.					
5. Pleasure - Degree to which a user obtains pleasure from fulfilling their personal needs.					
6. Comfort - Degree to which the user is satisfied with physical comfort.					
Freedom from Risk					
7. Economic Risk Mitigation - Degree to which a product or system mitigates the potential risk to financial status, efficient operation, commercial property, reputation or other resources in the intended contexts of use.					
8. Health and Safety Risk Mitigation - Degree to which a product or system mitigates the potential risk to people in the intended contexts of use.					
9. Environmental Risk Mitigation - Degree to which a product or system mitigates the potential risk to property or the environment in the intended contexts of use.					
Context coverage					
10. Context Completeness - Degree to which a product or system can be used with effectiveness, efficiency, freedom from risk, and satisfaction in all the specified contexts of use.					
11. Flexibility - Degree to which a product or system can be used with effectiveness, efficiency, freedom from risk and satisfaction in contexts beyond those initially specified in the requirements.					

Adopted from the International Organization for Standardization (ISO) 25010 for quality in use

Remarks/Comments/Suggestions:

Signature of Respondent

Appendix Table 5. Summarized background of the respondents - Information Technology experts

NO.	YEARS OF EXPERIENCE	CURRENT DESIGNATION	COMPANY
1	10 years	Full-Stack Developer	Office of the Information, Technology Office, GMA, Cavite
2	7 years	Network Engineer	Office of the Information, Technology Office, GMA, Cavite
3	5 years	IT Developer	Office of the Information, Technology Office, GMA, Cavite
4	6 years	Full-Stack Developer	Office of the Information, Technology Office, GMA, Cavite
5	5 years	Software Engineer	Office of the Information, Technology Office, GMA, Cavite

Appendix Table 6. Summarized background of the respondents – Human Resources

NO.	YEARS OF EXPERIENCE	CURRENT DESIGNATION	COMPANY
1	10 years	Human Resources	Toplis Solutions Inc. Silang
2	7 years	Human Resources	Toplis Solutions Inc. Silang
3	5 years	Human Resources	Toplis Solutions Inc. Silang

Appendix Table 7. Summarized background of the respondents – Administrator, supervisor, team leader, case picker, progress helper, receiving, RT Operator

NO.	YEARS OF EXPERIENCE	CURRENT DESIGNATION	COMPANY
1	2 years	Admin	Toplis Solutions Inc. Silang
2	3 years	Admin	Toplis Solutions Inc. Silang
3	1 year	Case Picker	Toplis Solutions Inc. Silang
4	3 years	Supervisor	Toplis Solutions Inc. Silang
5	3 years	Receiving	Toplis Solutions Inc. Silang
6	5 years	RT Operator	Toplis Solutions Inc. Silang

Appendix Table 7. Continued

NO.	YEARS OF EXPERIENCE	CURRENT DESIGNATION	COMPANY
7	4 years	Team Leader	Toplis Solutions Inc. Silang
8	3 years	Team Leader	Toplis Solutions Inc. Silang
9	3 years	Team Leader	Toplis Solutions Inc. Silang
10	3 years	RT Operator	Toplis Solutions Inc. Silang
11	2 years	RT Operator	Toplis Solutions Inc. Silang
12	2 years	RT Operator	Toplis Solutions Inc. Silang
13	2 years	RT Operator	Toplis Solutions Inc. Silang
14	1 year	RT Operator	Toplis Solutions Inc. Silang
15	5 years	Supervisor	Toplis Solutions Inc. Silang
16	5 years	Supervisor	Toplis Solutions Inc. Silang
17	5 years	Supervisor	Toplis Solutions Inc. Silang
18	3 years	Team Leader	Toplis Solutions Inc. Silang
19	3 years	Team Leader	Toplis Solutions Inc. Silang
20	4 years	Team Leader	Toplis Solutions Inc. Silang
21	4 years	Team Leader	Toplis Solutions Inc. Silang
22	5 years	Team Leader	Toplis Solutions Inc. Silang
23	1 year	Progress Helper	Toplis Solutions Inc. Silang
24	2 years	Progress Helper	Toplis Solutions Inc. Silang
25	1 year	Progress Helper	Toplis Solutions Inc. Silang
26	3 years	Progress Helper	Toplis Solutions Inc. Silang

Appendix Table 7. Continued

NO.	YEARS OF EXPERIENCE	CURRENT DESIGNATION	COMPANY
27	4 years	Progress Helper	Toplis Solutions Inc. Silang
28	3 years	Progress Helper	Toplis Solutions Inc. Silang
29	3 years	Progress Helper	Toplis Solutions Inc. Silang
30	5 years	Progress Helper	Toplis Solutions Inc. Silang
31	4 years	Progress Helper	Toplis Solutions Inc. Silang
32	3 years	Progress Helper	Toplis Solutions Inc. Silang
33	2 years	Progress Helper	Toplis Solutions Inc. Silang
34	1 year	Receiving	Toplis Solutions Inc. Silang
35	3 years	Receiving	Toplis Solutions Inc. Silang
36	1 year	Case Picker	Toplis Solutions Inc. Silang
37	2 years	Case Picker	Toplis Solutions Inc. Silang
38	3 years	Receiving	Toplis Solutions Inc. Silang
39	1 year	Case Picker	Toplis Solutions Inc. Silang
40	2 years	Receiving	Toplis Solutions Inc. Silang
41	2 years	Receiving	Toplis Solutions Inc. Silang
42	3 years	Case Picker	Toplis Solutions Inc. Silang
43	4 years	Receiving	Toplis Solutions Inc. Silang
44	5 years	Receiving	Toplis Solutions Inc. Silang
45	5 years	Case Picker	Toplis Solutions Inc. Silang
46	4 years	Case Picker	Toplis Solutions Inc. Silang

Appendix Table 7. Continued

NO.	YEARS OF EXPERIENCE	CURRENT DESIGNATION	COMPANY
47	4 years	Case Picker	Toplis Solutions Inc. Silang
48	4 years	Case Picker	Toplis Solutions Inc. Silang
49	5 years	Case Picker	Toplis Solutions Inc. Silang
50	3 years	Case Picker	Toplis Solutions Inc. Silang
51	3 years	Receiving	Toplis Solutions Inc. Silang
52	2 years	Receiving	Toplis Solutions Inc. Silang
53	3 years	Case Picker	Toplis Solutions Inc. Silang
54	4 years	Case Picker	Toplis Solutions Inc. Silang
55	4 years	Case Picker	Toplis Solutions Inc. Silang
56	3 years	Progress Helper	Toplis Solutions Inc. Silang
57	3 years	Progress Helper	Toplis Solutions Inc. Silang
58	3 years	Case Picker	Toplis Solutions Inc. Silang
59	4 years	Case Picker	Toplis Solutions Inc. Silang
60	4 years	Progress Helper	Toplis Solutions Inc. Silang
61	3 years	Progress Helper	Toplis Solutions Inc. Silang
62	5 years	Case Picker	Toplis Solutions Inc. Silang
63	2 years	Case Picker	Toplis Solutions Inc. Silang
64	5 years	Progress Helper	Toplis Solutions Inc. Silang
65	3 years	Case Picker	Toplis Solutions Inc. Silang
66	4 years	Case Picker	Toplis Solutions Inc. Silang

Appendix Table 7. Continued

NO.	YEARS OF EXPERIENCE	CURRENT DESIGNATION	COMPANY
67	5 years	Case Picker	Toplis Solutions Inc. Silang
68	4 years	Progress Helper	Toplis Solutions Inc. Silang
69	3 years	Progress Helper	Toplis Solutions Inc. Silang
70	3 years	Progress Helper	Toplis Solutions Inc. Silang
71	2 years	Case Picker	Toplis Solutions Inc. Silang
72	2 years	Case Picker	Toplis Solutions Inc. Silang